AI-Empowered SAFe®

Leading SAFe®

Thriving in the Digital Age with Business Agility

2026.02.24

Workbook

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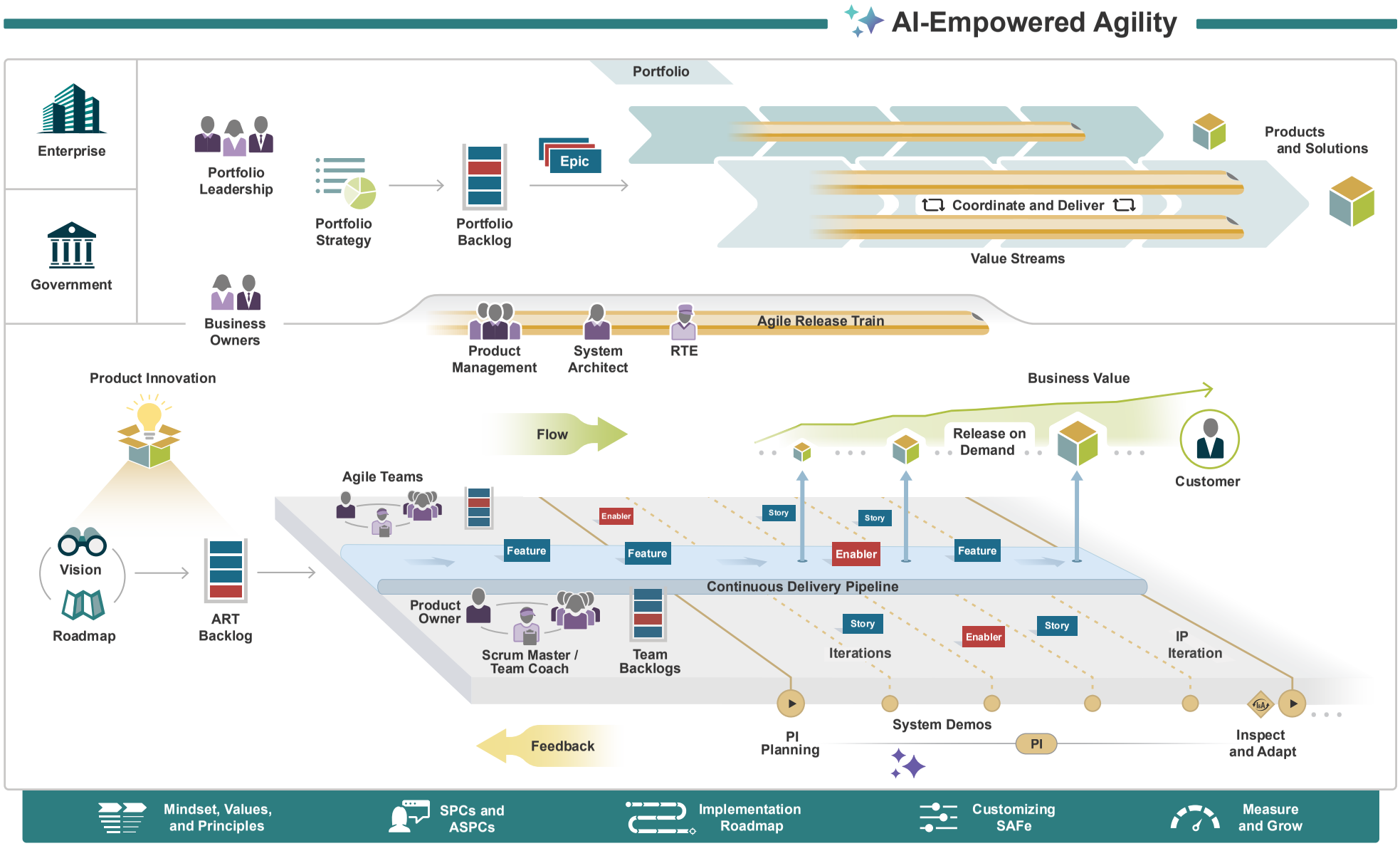
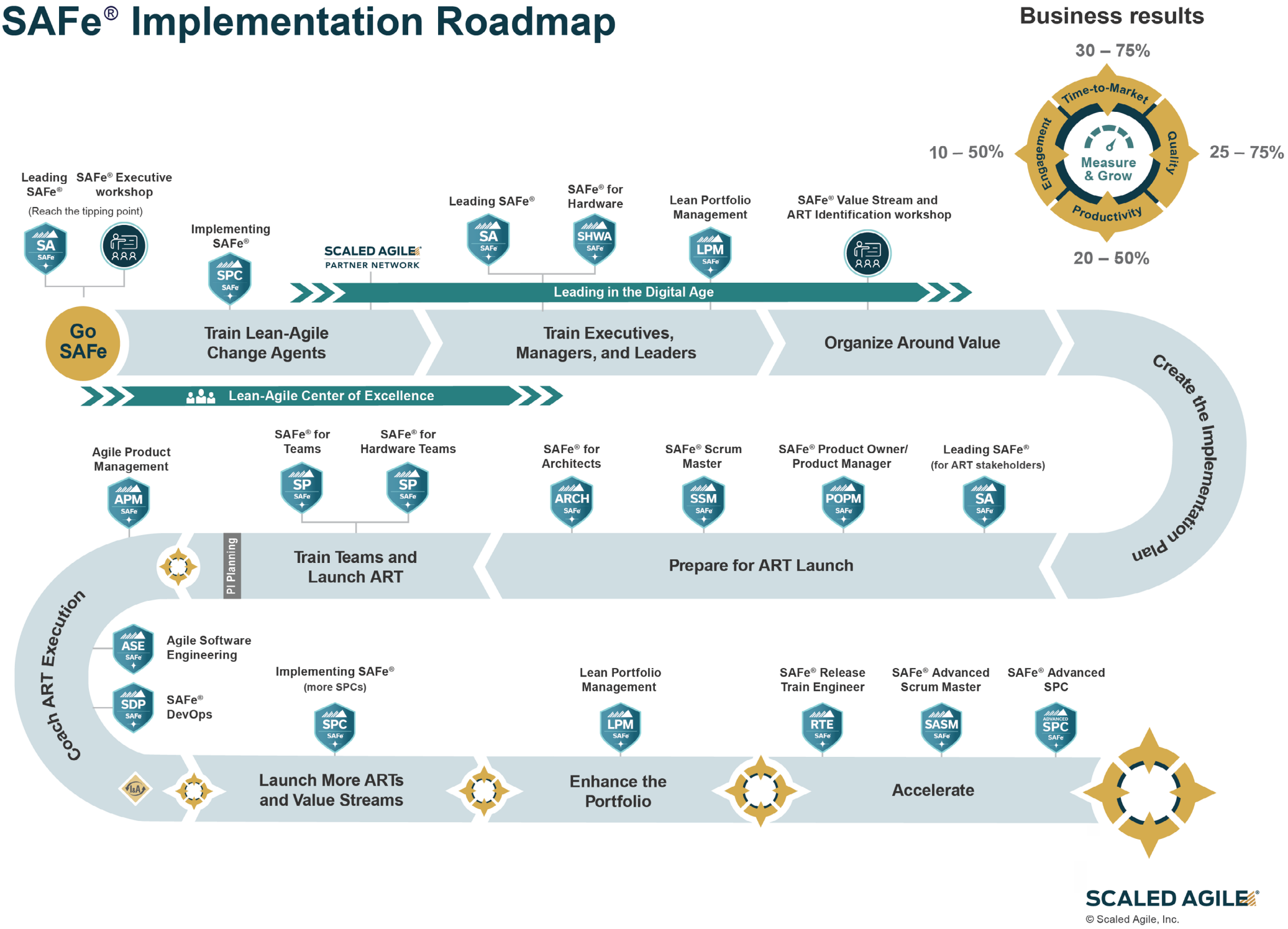
SAFe® Big Picture



SAFe® Implementation Roadmap



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AI-Empowered SAFe®

Leading SAFe

Adapting and thriving with the Scaled Agile Framework

SAFe® Course - Attending this course gives learners access to the SAFe Agilist exam and related preparation materials.

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Logistics

- Course meeting times
- Breaks
- Facilities
- Technology requirements
- Working agreements

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Discussion: Introductions

Duration: 5 min

In your group, go around and introduce yourselves. Share the following details:

- Your name
- Your role
- Your experience with SAFe
- What you hope to get out of this class

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Course outline

Leading SAFe

- Lesson 1: Adapting and Thriving with SAFe
- Lesson 2: Building a Foundation with Mindset, Values, and Principles
- Lesson 3: Team and Technical Agility
- Lesson 4: Product Development Flow
- Lesson 5: Lean Portfolio Management
- Lesson 6: Leading the Change
- Lesson 7: Get Certified

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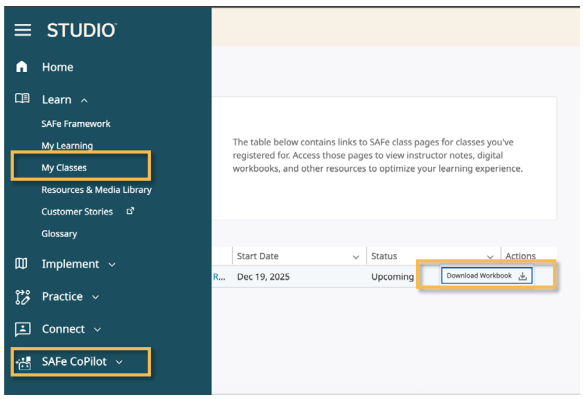
Activity: Find your SAFe Studio resources

Duration: 7 min

- **Step 1:** Navigate to My Classes on SAFe Studio
- **Step 2:** Click to download the Leading SAFe workbook
- **Step 3:** Navigate to the “During Class” tab and launch SAFe CoPilot
- **Step 4:** Ask SAFe CoPilot to give you a brief overview of SAFe and review the response



Launch the Class Page to download the workbook:
https://community.scaledagile.com/s/vc-class-list

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Lesson 1

Adapting and Thriving with SAFe

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Lesson Topics

1.1 Thriving in the age of disruption

1.2 Building a Lean-Agile organization

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Learning objectives

At the end of this lesson, you should be able to:

- Describe what is necessary to thrive in the face of constant disruption
- Summarize the elements of the Lean-Agile organization

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1.1 Thriving in the age of disruption

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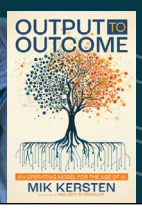
"When knowledge output becomes abundant, organizational design—not effort—will determine those who thrive in the Age of AI."

—Mik Kersten, *Output to Outcome*



Output to Outcome and *Project to Product* by Mik Kersten.
Portrait of Mik Kersten. Photo used with permission from Mik Kersten.

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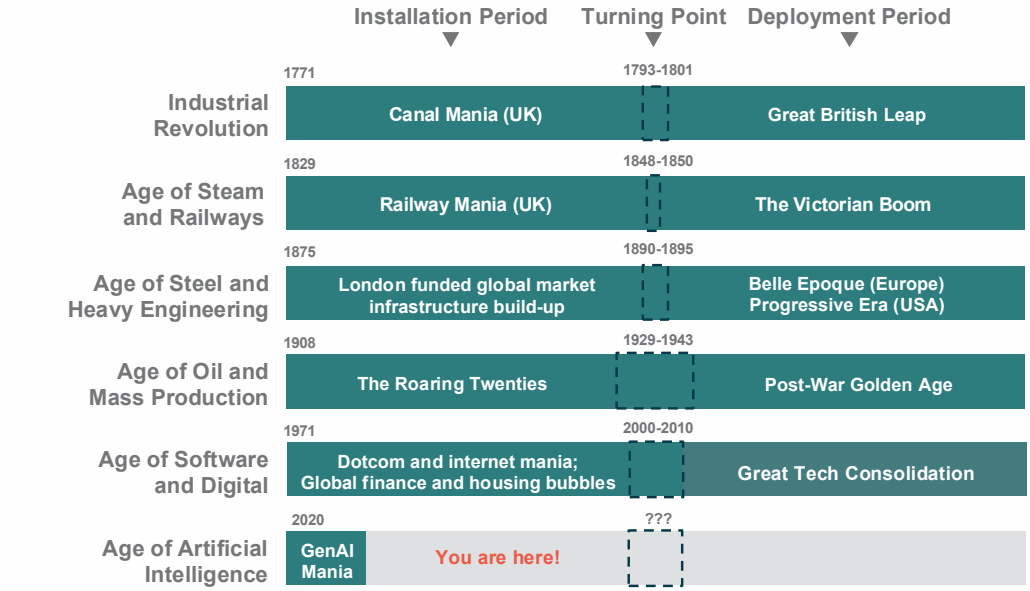
Technological revolutions change society



Source. The five technological revolutions, adapted from *Technological Revolutions and Financial Capital* by Carlota Perez (2002).

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Video: Welcome to the AI Age

Duration: 2 min

https://share.vidyard.com/watch/RiwC9z9oSF9whCXYeEnG43

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Discussion: AI disruption

Prepare: 7 min
Share: 3 min

- **Step 1:** In your group, explore how the widescale adoption of AI may disrupt how organizations operate and deliver value. Consider:
  - **Individuals and roles:** How might AI change skills, decision-making, or day-to-day work?
  - **Teams:** What opportunities or risks could AI introduce for flow, quality, and collaboration?
  - **The enterprise:** How might AI influence strategy, governance, or investment decisions?
- **Step 2:** Be prepared to share with the class.

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1.2 Building a Lean-Agile organization

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Rethinking the organization

"The world is now changing at a rate at which the basic systems, structures, and cultures built over the past century cannot keep up with the demands being placed on them."

—John P. Kotter, *Accelerate*

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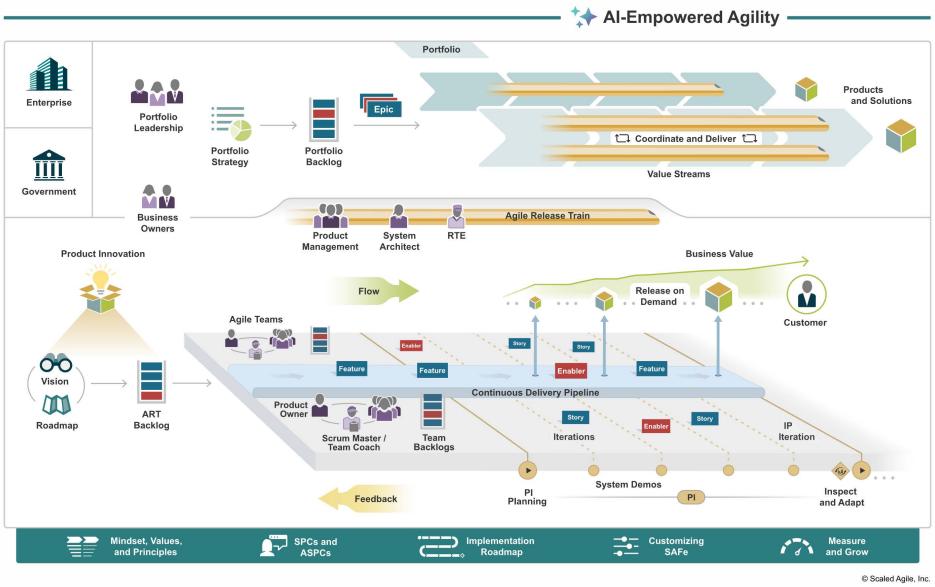
The SAFe Big Picture



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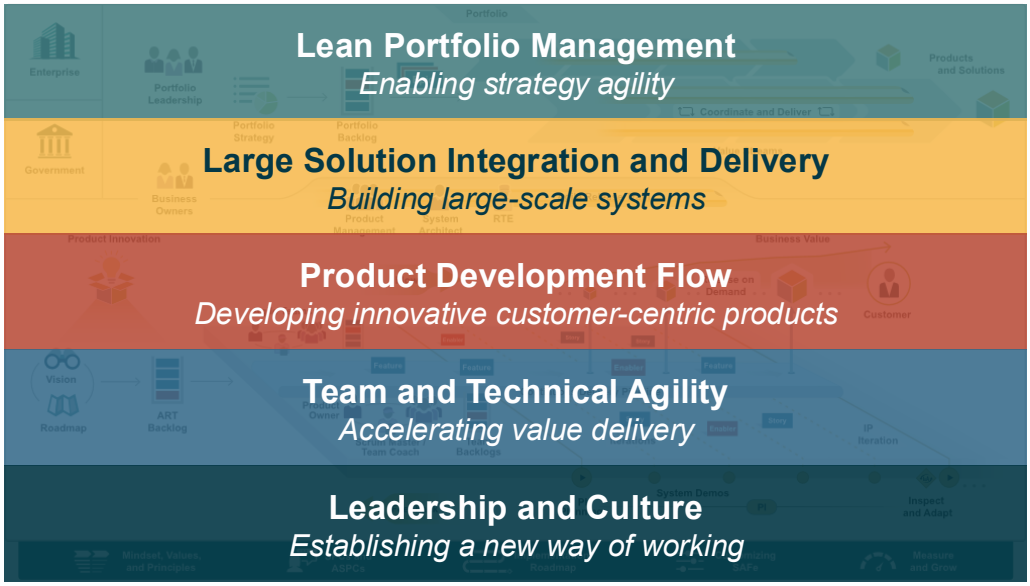
The five disciplines of the Lean-Agile organization



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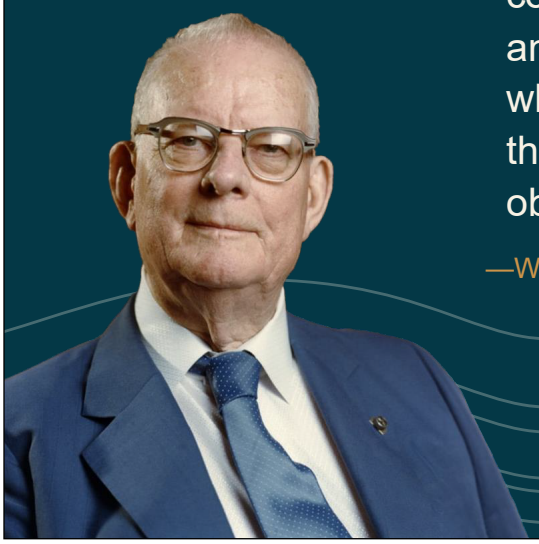
The management challenge



"It is not enough that top management commit themselves for life to quality and productivity. They must know what it is that they are committed to—that is, what they must do. These obligations can not be delegated."

—W. Edwards Deming, *Out of the Crisis*

Portrait of W. Edwards Deming.
Photo courtesy of The W. Edwards Deming Institute®.

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Leaders are critical in SAFe

By driving the transformation, leaders create momentum for SAFe by:

- Sponsoring the transformation, leading the change, and creating a compelling vision for business agility
- Modeling the SAFe mindset and principles; empowering decentralized decision-making
- Investing in people: coaching, mentoring, upskilling, and career development
- Creating an environment that enables people to plan, align, collaborate, and improve flow

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SAFe Practice Consultants lead the transformation

- Experienced change agents who implement SAFe and accelerate organizational agility at scale
- Guide the transformation, teach SAFe, coach for better flow, and help leaders and teams adopt Lean-Agile ways of working
- Build the guiding coalition, run workshops, and enable executive sponsorship
- Help identify value streams and launch ARTs, aligning strategy with execution

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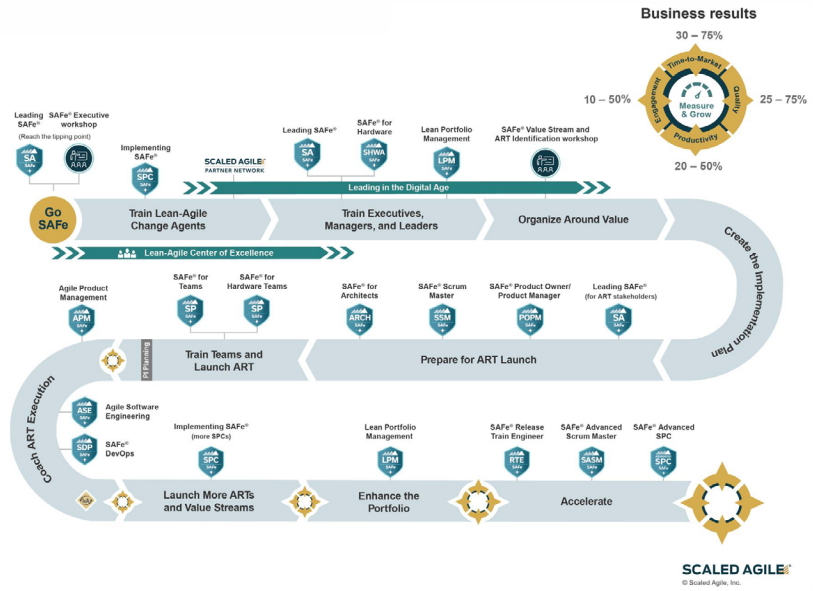
SAFe Implementation Roadmap



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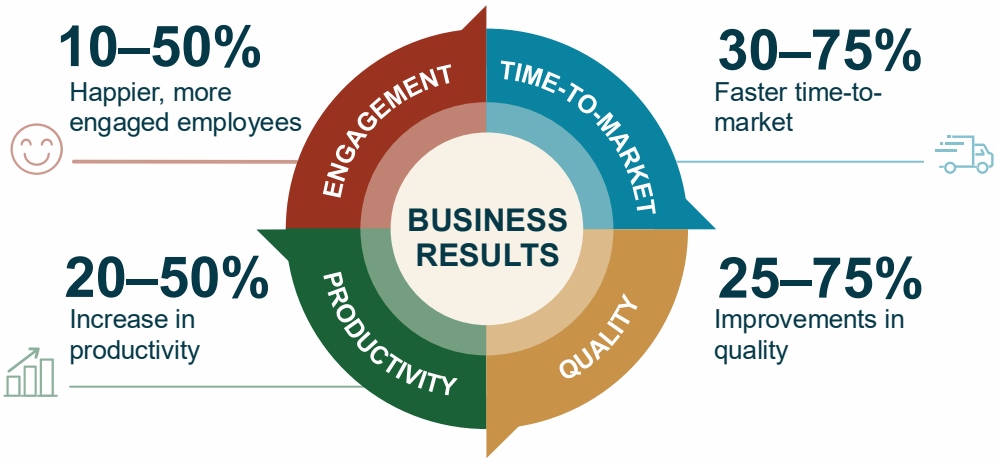
Why SAFe?

SAFe's business benefits are derived directly from case studies written by SAFe customers.



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1-22

Discussion: What are you hoping to get from SAFe?

Prepare: 5 min
Share: 3 min

- **Step 1:** Imagine it's one year from today, and that your organization has been working with SAFe for a year
- **Step 2**: In your group, discuss the following questions:
  - What challenges do you hope SAFe will help unlock by then?
  - What feels difficult, slow, or frustrating today that you want to see change?
  - If these challenges were unlocked, what would be noticeably different in how work gets done?
  - What other indicators would tell you that SAFe is improving outcomes
- **Step 3:** Be prepared to share with the class

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The ability to solve real business problems is the best measure of SAFe success

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Lesson review

In this lesson, you:

- Described what is necessary to thrive in face of constant disruption
- Summarized the elements of the Lean-Agile organization

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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson.

"SAFe Practice Consultant (SPC) and Advanced SAFe Practice Consultant (ASPC),"
https://framework.scaledagile.com/spc/

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Deming, W. Edwards. *Out of the Crisis.* Cambridge: The MIT Press, 2000. 23.

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Kersten, Mik. *Project to Product: How to Survive and Thrive in the Age of Digital Disruption with the Flow Framework.* IT Revolution: Portland, 2018. Cover.

Kotter, John P. *Accelerate: Building Strategic Agility for a Faster-Moving World.* Boston: Harvard Business Review Press, 2014. vii, viii.

Perez, Carlota. *Technological Revolutions and Financial Capital: The Dynamics of Bubbles and Golden Ages.* Cheltenham: Edward Elgar Publishing Limited, 2002. 107.

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Lesson 2

Building a Foundation with Mindset, Values, and Principles

SAFe® Course - Attending this course gives learners access to the SAFe Agilist exam and related preparation materials.

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Lesson Topics

2.1 The Lean-Agile mindset and SAFe Core Values

2.2 Applying SAFe Principles

2.3 Empowering agility with AI

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Learning objectives

At the end of this lesson, you should be able to:

- Describe the value of a Lean-Agile mindset
- Recognize the SAFe Core Values
- Apply the SAFe Lean-Agile principles
- Explain the differences between various types of AI models
- Describe ways to ensure responsible AI usage

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2.1 The Lean-Agile mindset and SAFe Core Values

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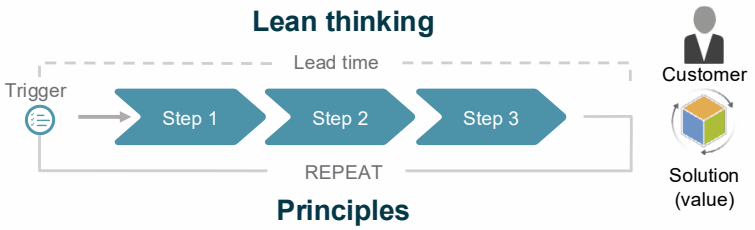
Lean thinking



Principles

1. Precisely specify value by product
2. Identify the value stream for each product
3. Make value flow without interruptions
4. Let the customer pull value from the producer
5. Pursue perfection

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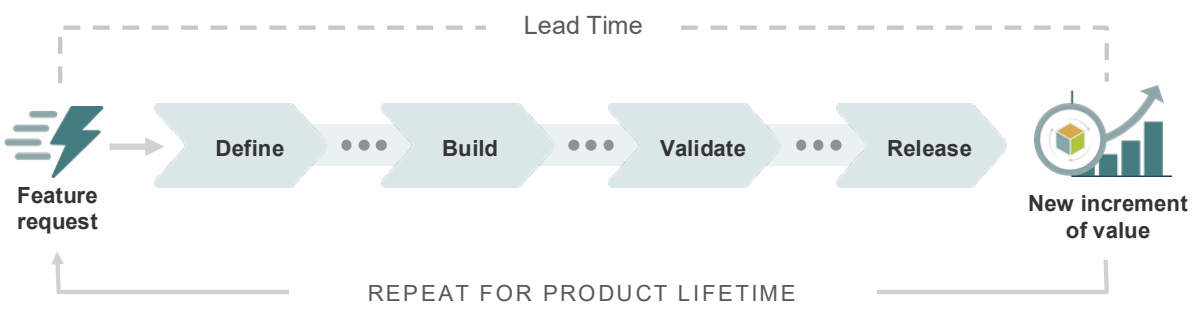
Identify the value stream for each product

Each value stream:

- Includes activities from recognizing an opportunity through release and validation
- Contains the steps, the flow of information and material, and the people who develop the solution



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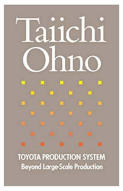
Optimize the full value stream

Toyota Motor Co., Ltd. Company History Editorial Committee, Public domain, via Wikimedia Commons



Toyota Production System by Taiichi Ohno.

"All we are doing is looking at the time line... from the moment the customer gives us an order to the point when we collect the cash. And we are reducing that time line by removing the non-value-added wastes."

—Taiichi Ohno, as quoted by Norman Bodek, from the foreword of *Toyota Production System*

- Most problems with process will surface as delays
- Most of the time spent getting to market is a result of these delays
- Reducing delays is the fastest way to reduce time-to-market

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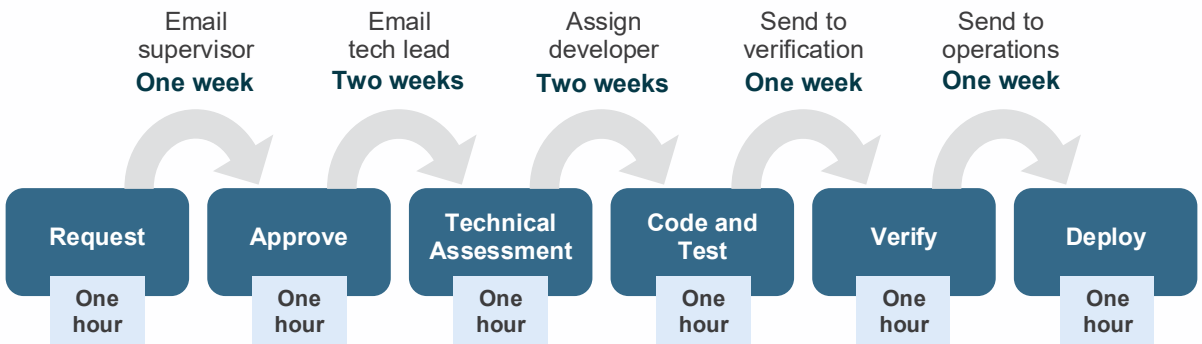
Focus on minimizing the delays



Six hours of value… **…delivered in seven weeks**

$$\frac{\text{1 day}}{\text{49 days}} = \text{2\% Flow Efficiency}$$

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Activity: Identify your value stream

Duration: 7 min

- **Step 1:** Individually, select a solution your organization is developing
- **Step 2:** Identify the customer you are building it for
- **Step 3:** In your workbook, sketch out the basic workflow steps for the value stream
- **Step 4:** Identify a few delays in the value stream and the potential causes

Note: You will build on your example value stream through the subsequent activities in this lesson.

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The Agile Manifesto

Agile values

We are uncovering better ways of developing software by doing it and helping others do it

Through this work we have come to value:

Individuals and interactions over processes and tools

Working software over comprehensive documentation

Customer collaboration over contract negotiation

Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

Reference: Agile Manifesto

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Identifying your value stream

Instructions: Select a Solution your Enterprise is currently developing and identify the Customer you are building it for. In the text boxes below, sketch out the steps in the Value Stream.

Solution:

Customer:

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Identifying your delays

Instructions: Identify three delays from the Value Stream and write them down. Write down what you think might be some potential causes for the delays. Be prepared to share with the class.

Delay #1:

Delay #1 cause:

Delay #2:

Delay #2 cause:

Delay 3:

Delay #3 cause:

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Activity: Agile principles at scale

Prepare: 5 min
Share: 5 min

- **Step 1:** As a group, select one of the principles behind the Agile Manifesto from the following slides listed in your workbook
- **Step 2:** Categorize the principle as:
  - Not applicable at scale
  - Works as is at scale
  - Requires rethinking for scale
- **Step 3:** Be prepared to share with the class

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Exemplifying SAFe Core Values

Alignment

- Communicate the vision, mission, and strategy
- Connect strategy to execution
- Speak with a common language
- Constantly check for understanding
- Understand your customer

Transparency

- Create a trust-based environment
- Communicate directly, openly, and honestly
- Turn mistakes into learning moments
- Visualize work
- Provide ready access to needed information

Respect for people

- Hold precious what it is to be human
- Value diversity of people and opinions
- Grow people through coaching and mentoring
- Embrace 'your customer is whoever consumes your work'
- Build long-term partnerships based on mutual benefit

Relentless improvement

- Create a constant sense of urgency
- Build a problem-solving culture
- Reflect and adapt frequently
- Let facts guide improvements
- Provide time and space for innovation

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Agile principle at scale

Instructions: Select one of the principles behind the Agile Manifesto from the list below. Categorize the principle as 'Not Applicable,' 'Works as-is' and 'Requires rethinking at scale.'

Principles	Works as-is	Requires rethinking at scale	Not applicable
1. Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.	○	○	○
2. Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.	○	○	○
3. Deliver working software frequently, from a couple of weeks to a couple of months, with a preference for the shorter timescale.	○	○	○
4. Business people and developers must work together daily throughout the project.	○	○	○
5. Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.	○	○	○
6. The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.	○	○	○
7. Working software is the primary measure of progress.	○	○	○
8. Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.	○	○	○
9. Continuous attention to technical excellence and good design enhances agility.	○	○	○
10. Simplicity–the art of maximizing the amount of work not done–is essential.	○	○	○
11. The best architectures, requirements, and designs emerge from self-organizing teams.	○	○	○
12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.	○	○	○

Notes

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2.2 Applying SAFe Principles

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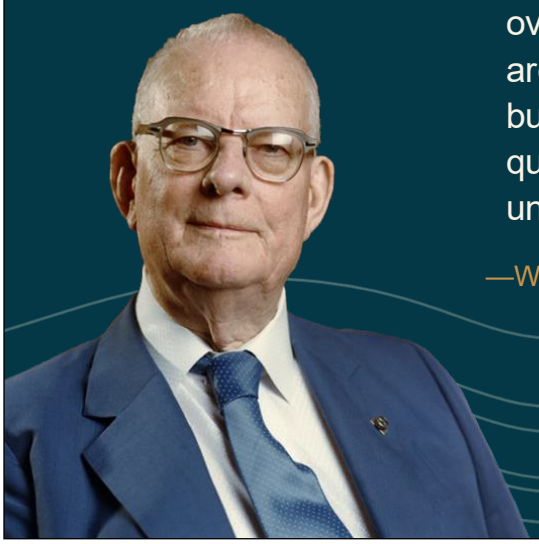
Why focus on the principles?

“ A common disease that afflicts management and government administration the world over is the impression that ‘Our problems are different.’ They are different, to be sure, but the principles that will help to improve quality of product and of service are universal in nature.”

—W. Edwards Deming, *Out of the Crisis*



Portrait of W. Edwards Deming.
Photo courtesy of The W. Edwards Deming Institute®.

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SAFe Lean-Agile principles

#1 Take an economic view

#2 Apply systems thinking

#3 Assume variability; preserve options

#4 Build incrementally with fast, integrated learning cycles

#5 Base milestones on objective evaluation of working systems

#6 Make value flow without interruptions

#7 Apply cadence, synchronize with cross-domain planning

#8 Unlock the intrinsic motivation of knowledge workers

#9 Decentralize decision-making

#10 Organize around value

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#1 Take an economic view

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Video: SAFe Principle #1: Take an economic view

Duration: 2 min

https://share.vidyard.com/watch/UzsPSxPPBgJrVPhrvX4Htg

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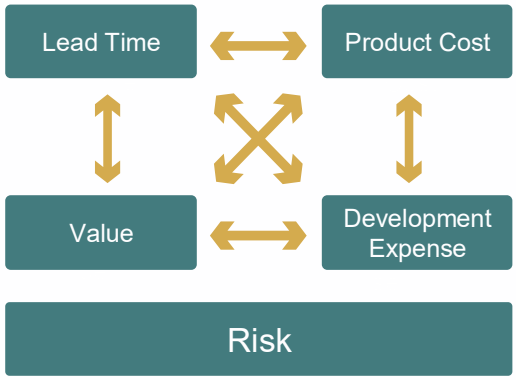
Economic trade-offs

Understanding trade-off parameters:

- Sequence jobs for maximum benefit
- Do not consider money already spent
- Make economic choices continuously
- Empower local decision making
- If you only quantify one thing, quantify the cost of delay



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#2 Apply systems thinking

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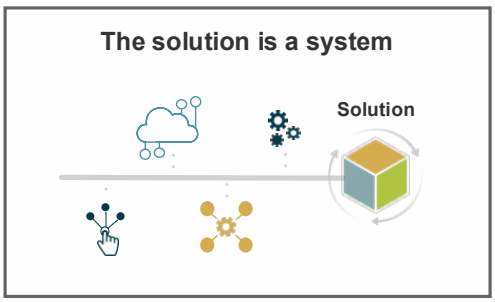
Systems thinking



Attributes of systems thinking

- Optimizing a component does not optimize the system
- For the system to behave well as a system, a higher-level understanding of behavior and architecture is required
- The value of a system passes through its interconnections
- A system can evolve no faster than its slowest integration point

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#3 Assume variability; preserve options

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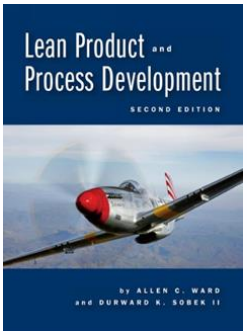
Development occurs in an uncertain world



Lean Product and Process Development by Allen C. Ward and Durward K. Sobek II

"Rather than try to pick an early winner, aggressively eliminate alternatives through analysis, simulation, and testing."

—Allen C. Ward, *Lean Product and Process Development*

- It's impossible to know everything at the start
- Requirements must be flexible to make economic design choices
- Designs must be flexible to support changing requirements
- Preservation of options improves economic results

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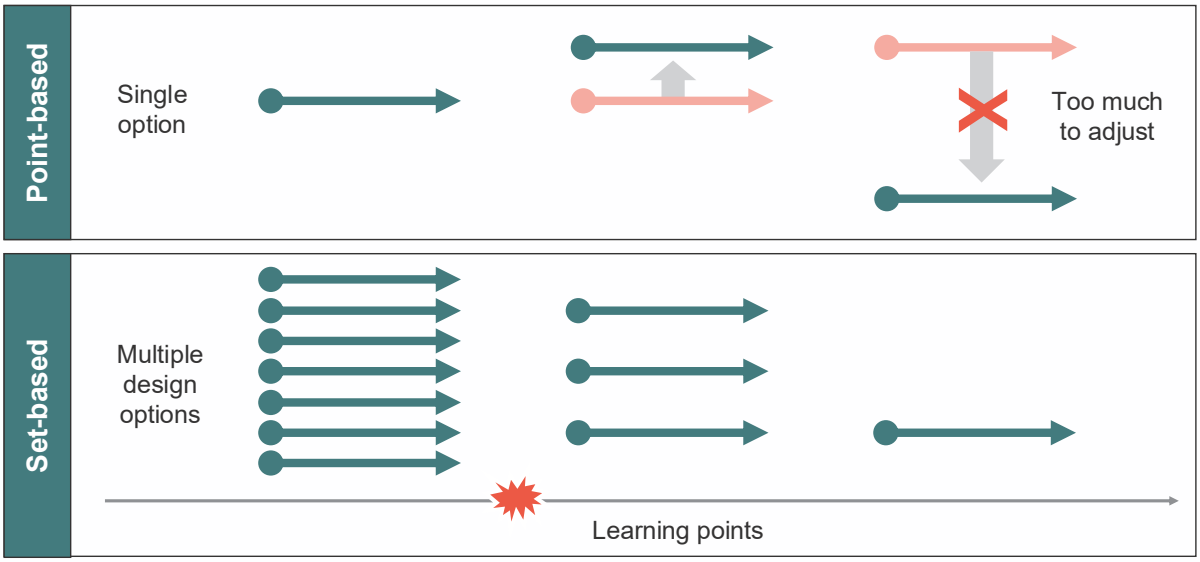
Apply a set-based approach



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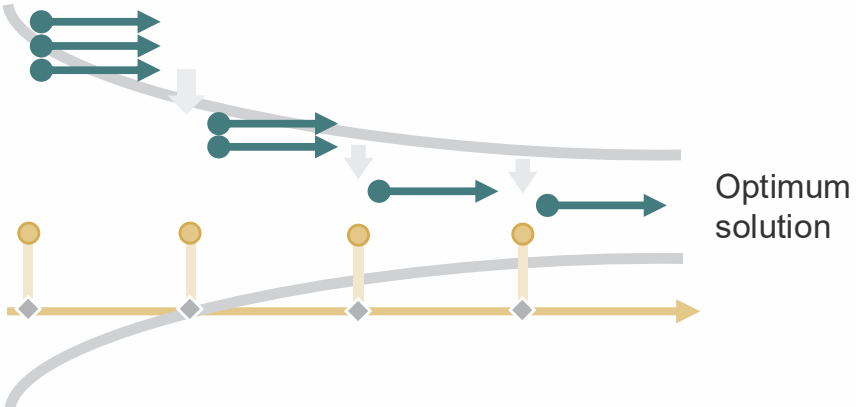
Iterate to the optimum solution

Set-Based Design facilitates learning and allows for continuous, cost-effective adjustments towards an optimum solution.



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Activity: Set-Based Design in context

Duration: 7 min

- **Step 1:** Individually, identify an upcoming technical or business milestone in the value stream you previously identified
- **Step 2:** In your workbook, identify some design options that might help to better ensure you can meet that milestone

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#4 Build incrementally with fast, integrated learning cycles

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Set-Based Design in context

Instructions: Identify an upcoming technical or business Milestone in the Value Stream you previously identified. In the notes section below, identify some design options that might help to better assure you can meet that Milestone.

Notes

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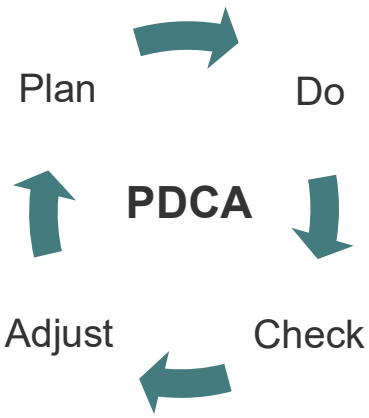
Apply fast learning cycles

Fast feedback accelerates knowledge.

- Improves learning efficiency by decreasing the time between action and effect
- Reduces the cost of risk-taking by truncating unsuccessful paths quickly
- Is facilitated by small batch sizes
- Requires increased investment in development environment

The shorter the cycles, the faster the learning.

The iterative learning cycle



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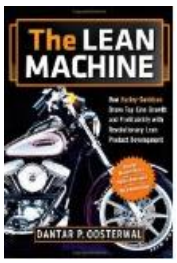
Integration points control product development

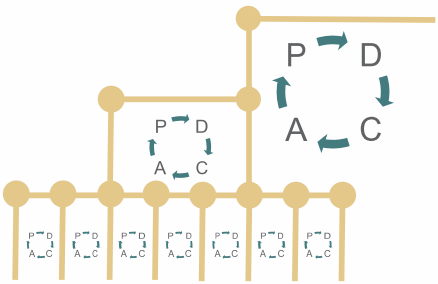


"Product development is the creation of reusable knowledge through set-based design and the establishment of development cadence and flow."

—Dantar P. Oosterwal, *The Lean Machine*

The Lean Machine
by Dantar P. Oosterwal

- Integration points accelerate learning
- Development can proceed no faster than the slowest learning loop
- Improvement comes through synchronization of design loops and faster learning cycles



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Activity: Accelerating learning cycles in context

Duration: 5 min

- **Step 1:** Individually, consider the value stream and solution that you previously identified
- **Step 2:** In your workbook, identify which integration points might apply to reduce the risk
- **Step 3:** Be prepared to share with the class

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#5 Base milestones on objective evaluation of working systems

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Instructions: Consider the Value Stream and Solution that you previously identified. In the notes section below, identify which integration points might apply to reduce the risk of being in the zone of 'false positive feasibility.'

Notes

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The problem of phase-gate milestones

"There was in fact no correlation between exiting phase gates on time and the success of a project.… The data even suggested that the inverse may be true."

—Dantar P. Oosterwal, *The Lean Machine*

- They force design decisions too early; this encourages false positive feasibility
- They assume a 'point' solution exists and can be built correctly the first time
- They create huge batches and long queues, and they centralize requirements and design in program management

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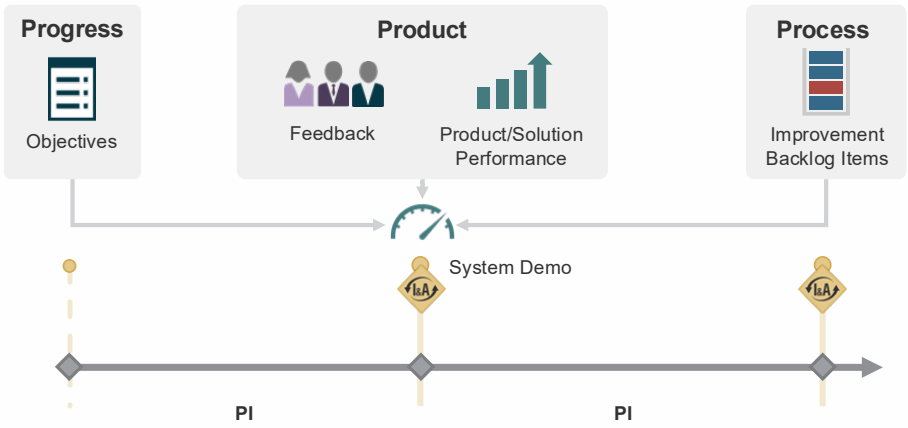
Apply objective milestones

Planning Interval (PI) System Demos are orchestrated to deliver objective progress, product, and process metrics.



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#6 Make value flow without interruptions

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Video: SAFe Principle #6: Make value flow without interruptions

Duration: 7 min

https://share.vidyard.com/watch/79AdUyJhpdiF9WMPmocQvY

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Activity: Experience a large batch size

Duration: 5 min

- **Step 1:** Create groups of five people with 10 coins per group. Designate one person as the timekeeper. The remaining four people will be processing the coins.
- **Step 2:** Process each coin person by person.
- **Step 3:** Pass all coins at the same time to the next person, who repeats step two until all four people are done.
- **Step 4:** The timekeeper stops the timer and records the total time.

https://share.vidyard.com/watch/xHfsetxaqmSuuFCzfMJrty

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Activity: Experience a small batch size

Duration: 5 min

- **Step 1:** Ensure that the timekeeper is ready to start the timer
- **Step 2:** This time, each person in their group processes one coin at a time and immediately passes each coin to the next person
- **Step 3:** The timekeeper will stop the timer when the last person flips the last coin and records the result

Optional video demonstrating this exercise: part two of three

https://share.vidyard.com/watch/QSixQvUJxD9uBWMQB5asNj

Optional video demonstrating this exercise: part three of three

https://share.vidyard.com/watch/zcBpJ7uuiyZJ5nS8wgqhRk

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#7 Apply cadence, synchronize with cross-domain planning

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Cadence and synchronization

Cadence

- Converts unpredictable events into predictable occurrences and lowers cost
- Makes waiting times for new work predictable
- Supports regular planning and cross-functional coordination
- Limits batch sizes to a single interval
- Controls injection of new work
- Provides scheduled integration points

Note: Delivering on cadence requires scope or capacity margin

Synchronization

- Causes multiple events to happen simultaneously
- Facilitates cross-functional trade-offs
- Provides routine dependency management
- Supports full-system integration and assessment
- Provides multiple feedback perspectives

Note: To work effectively, design cycles must be synchronized

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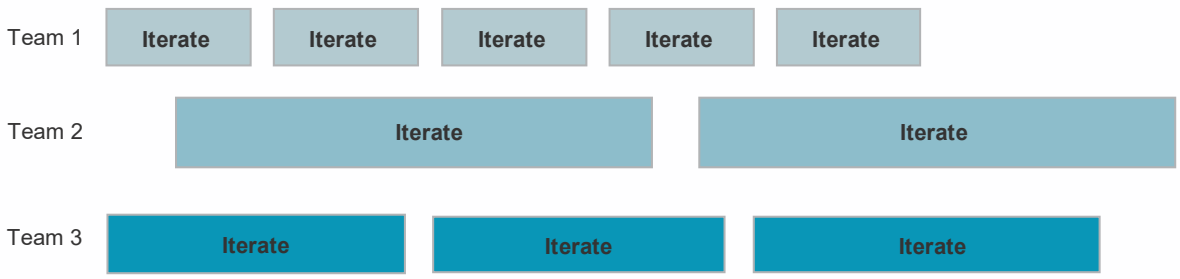
Cadence without synchronization is not enough



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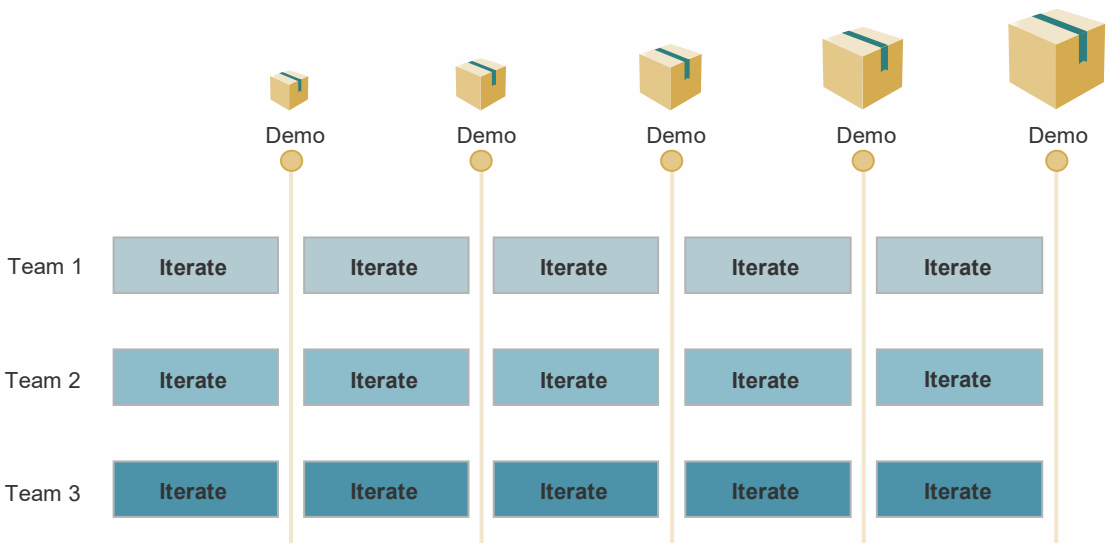
Synchronize to assure delivery



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#8 Unlock the intrinsic motivation of knowledge workers

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On managing knowledge workers

"That [knowledge workers] know more about their job than anybody else in the organization is part of the definition of knowledge workers."

—Peter Drucker, *The Essential Drucker*

- Workers themselves are most qualified to make decisions about how to perform their work
- The workers must be heard and respected for management to lead effectively
- Knowledge workers must manage themselves and require autonomy
- Continuing innovation must be part of the work, the tasks, and the responsibilities of knowledge workers

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Unlocking intrinsic motivation with autonomy, mastery and purpose

- **Autonomy** is the desire to be self-directed and have control over what to work on, how to do the work, and who to work with
- **Mastery** is the urge to get better at what to do and improve personal and team skills
- **Purpose** is the desire to do something that matters and has meaning

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#9 Decentralize decision-making

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Video: MindSpring Presents: “Greatness” by David Marquet

Duration: 10 min

https://www.youtube.com/watch?v=OqmdLcyES_Q

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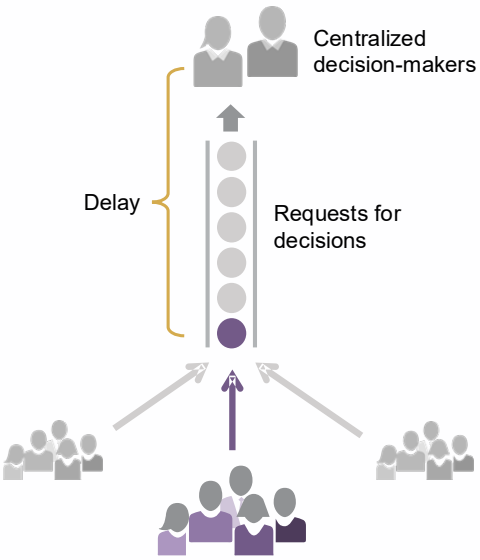
Excessive centralization of decisions leads to delays

- Openly discuss decision-making and explore moving authority closer to where work is done.
- Establish a decision-making framework that equips knowledge workers with the information to make good decisions.
- Provide clarity on organizational objectives, coach problem-solving, and offer opportunities to develop decision-making skills.
- Take responsibility for making and communicating strategic decisions. Decentralize all other decisions.



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Decentralize decision-making

Define the economic logic behind a decision; empower others to make the changes.

Centralize	Decentralize everything else
▶ **Infrequent:** Not made very often and usually not urgent (Example: Internationalization strategy)	▶ **Frequent:** Routine, everyday decisions (Example: Team and ART backlog)
▶ **Long-lasting:** Once made, highly unlikely to change (Example: Common technology platform)	▶ **Time critical:** High cost of delay (Example: Point release to customer)
▶ **Significant economies of scale:** Provide large and broad economic benefit (Example: Compensation strategy)	▶ **Requires local information:** Specific and local technology or customer context is required (Example: Feature criteria)

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Activity: Decentralize decision-making

Prepare: 5 min
Share: 2 min

- **Step 1:** Individually, consider three significant decisions you are currently facing. Write them in the table provided in your workbook.
- **Step 2:** Rate each decision based on the frequency, time criticality, and economies of scale, assigning a value of zero, one, or two.
- **Step 3:** Add the total values: zero to three = centralize; four to six = decentralize.

	Frequent? Y=2 N=0	Time-critical? Y=2 N=0	Economies of scale? Y=0 N=2	Total
Defining Strategic themes	0	1	0	1 (centralize)
Team Backlog refinement	2	2	2	6 (decentralize)
Define PI Dates	0	1	0	1 (centralize)

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Decentralize decision-making

Instructions: Consider three significant decisions you are currently facing. Write them in the table below. Rate each decision based on the frequency, time criticality, and economies of scale, assigning a value of zero, one, or two. Add the total values: zero to three = centralize; four to six = decentralize.

Decision	Frequent? Y=2 N=0	Time-critical Y=2 N=0	Economies of scale? Y=0 N=2	Total

Notes

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#10 Organize around value

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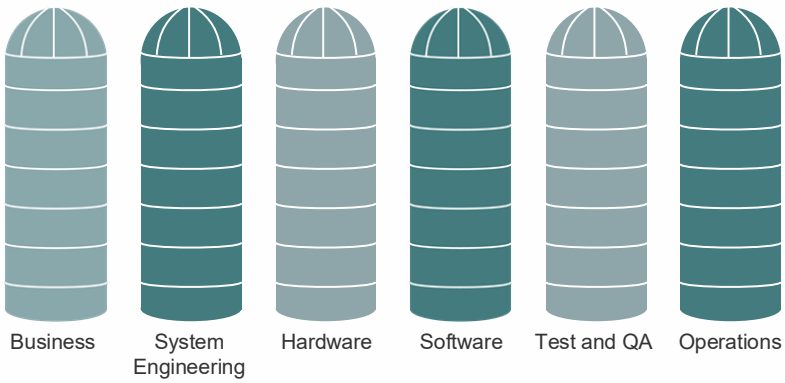
Value doesn't follow silos



Management challenge: connect the silos

- Value delivery is inhibited by handoffs and delays
- Political boundaries can prevent cooperation
- Silos encourage geographic distribution of functions
- Communication across silos is difficult

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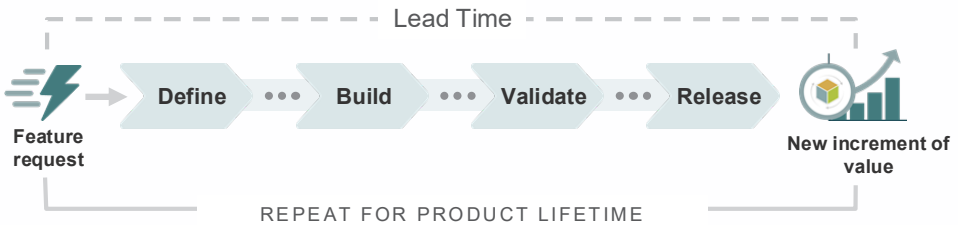
Instead, organize around development Value Streams

"Al Ward tells us that the aim of development is in fact the creation of profitable operational value streams."

—John Shook, from the foreword of *Lean Product and Process Development*

Each development Value Stream:

- Includes activities from recognizing an opportunity through release and validation
- Contains the steps, the flow of information and material, and the people who develop the solution used by the Operational Value Streams



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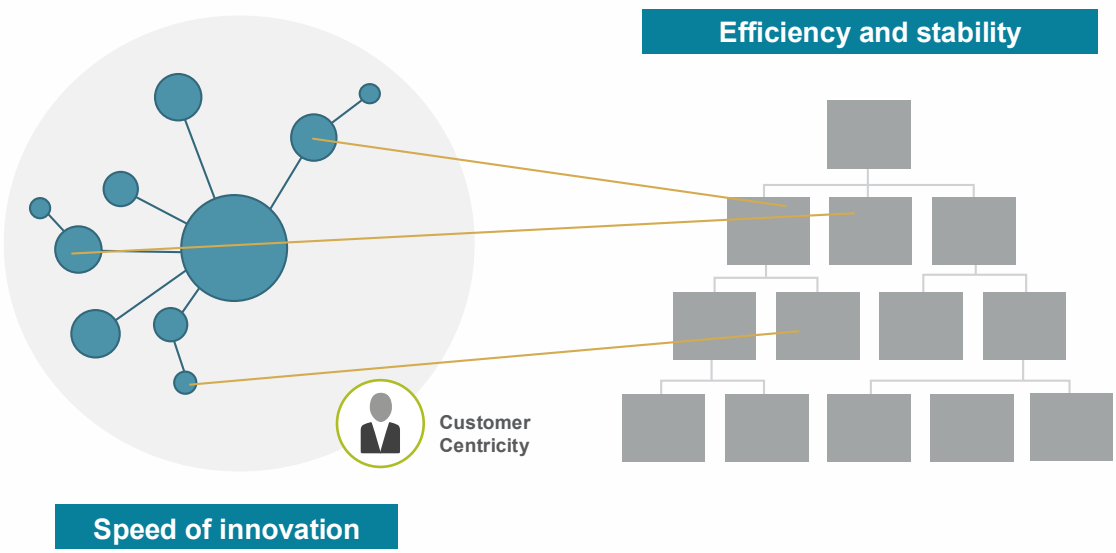
We need a dual operating system for Business Agility



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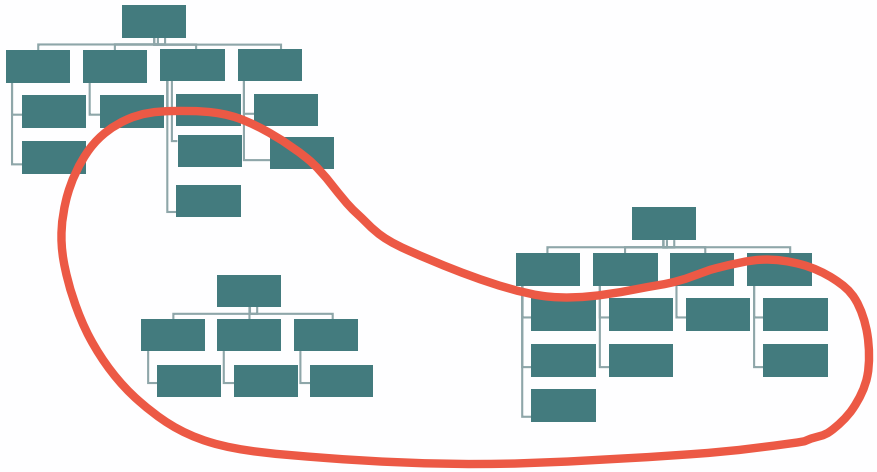
Value flows across organizational boundaries

Identify the Value Streams within which to build one or more Agile Release Trains.



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2.3 Empowering agility with AI

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AI changes the speed, not the foundations

- AI increases decision velocity, experimentation, and feedback
- Faster systems require clear values, decentralized decisions, and objective evaluation
- SAFe foundations help organizations:
  - Apply systems thinking to AI-enabled solutions
  - Build incrementally with fast learning cycles
  - Balance innovation with responsibility
  - Explore AI solutions that are aligned with the organization's strategy

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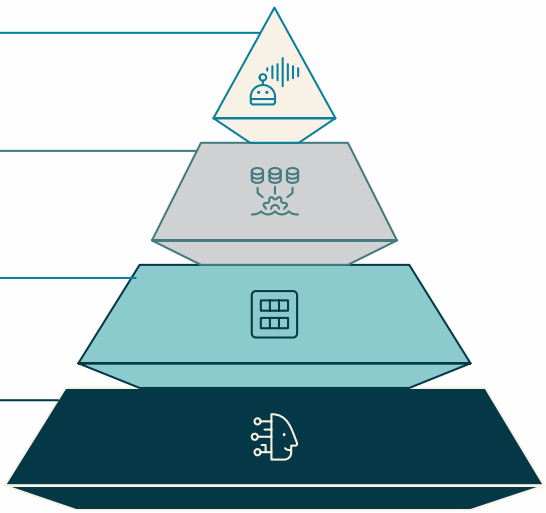
Understanding AI, ML, DL, and GenAI

Generative AI (GenAI)
DL creating new realistic outputs

Deep Learning (DL)
ML using multi-layered neural networks

Machine Learning (ML)
AI systems that learn from data patterns

Artificial Intelligence (AI)
Broad field of human-like machine intelligence



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Human vs. machine intelligence

Two different kinds of modern intelligence:

Humans

- Flexible and emotionally aware
- Capable of abstraction and moral reasoning
- Can understand context and nuance

does not replace

Machines

- Narrow and task-specific
- Excel at finding patterns in large datasets
- Execute those patterns at incredible scale
- Lack consciousness, intent, or ethical judgment

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Types of GenAI – Prompting and Large Language Models

- Large Language Models (LLMs) are AI systems trained on vast amounts of text so they can understand questions and generate human-like responses
  - "Prompting" is simply how users communicate with them (giving them instructions or questions)
- Best for:
  - Explaining concepts in plain language
  - Summarizing or rewriting content
  - Brainstorming ideas
  - Answering general questions
- Example: ChatGPT

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Types of GenAI – Retrieval Augmented Generation

- Retrieval-Augmented Generation (RAG) combines an LLM with trusted sources of information so responses are grounded in real, current data
  - Think of it as giving the AI a reference library
- Best for:
  - Answering questions using company documents
  - Reducing incorrect or made-up answers
  - Providing context-aware, up-to-date responses
- Example: A customer support chatbot that references knowledge base articles

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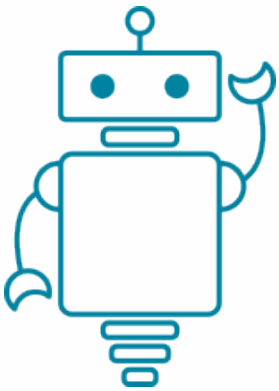
Types of GenAI – Agentic

- Agentic AI builds on LLMs and RAG to pursue a goal, take initiative, and coordinate multiple steps while maintaining accountability
- Best for:
  - Managing complex, ongoing work
  - Monitoring systems over time
  - Surfacing risks and trade-offs early
  - Reducing decision overhead
- Example: A work-intake assistant that watches incoming requests, then groups related work and flags overload or conflicts, and suggests prioritization



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AI is revolutionary, and not without risks

- An attorney filed a brief with hallucinated cases and citations obtained from ChatGPT in a U.S. District court.
- A healthcare AI used to identify patients needing “high-risk care management” was more likely to identify white patients over Black.
- A self-driving car manufacturer recalled its entire fleet of autonomous vehicles after a crash involving a pedestrian.
- A class-action lawsuit accused Google of breaking copyright laws when it created the datasets used to train its large language models.
- Samsung employees accidentally shared confidential information while using ChatGPT for help checking source code.

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Common AI risks

- **Bias:** AI systems can exhibit bias if they are trained on biased datasets. This can lead to unfair or discriminatory outcomes.
- **Hallucination:** AI creates outputs that seem wrong or make no sense because of mistakes in its data or how it was made.
- **Data leaks:** Sensitive or confidential information can be unintentionally exposed when using AI tools, especially those hosted by third-party vendors.

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Responsible AI use

Assume it's public

- **Never** input confidential company data, trade secrets, financial records, employee information, or client PII into a public AI tool
- Use only company-approved, private AI instances for sensitive work

Know the rules

- Understand and adhere to your company's specific guidelines on using AI tools.
- Which tools are approved? How should you disclose AI-generated content?
- When in doubt, ask.

Be the skeptic

- Treat AI-generated content as a first draft, not a final product.
- Always fact-check data, statistics, and critical claims before using them.
- Demand evidence from the AI. You are responsible for the final output.

Think critically

- Carefully review text and images for subtle biases, stereotypes, or any other language that could harm people, organizations, or society
- If the output feels "off" or unfair, challenge it and refine your prompt or approach

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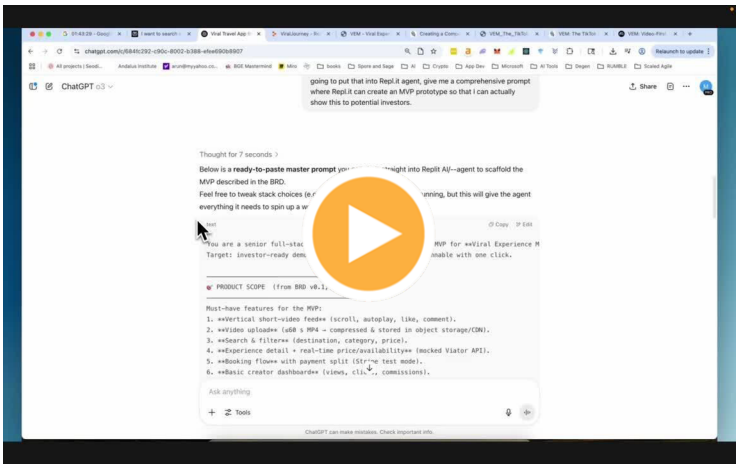
Video: Demo: The Art of the Possible with AI [Clip]

Duration: 5 min



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Lesson review

In this lesson, you:

- Described the value of a Lean-Agile mindset
- Recognized the SAFe Core Values
- Applied the SAFe Lean-Agile principles
- Explained the differences between various types of artificial intelligence (AI) models
- Described ways to ensure responsible AI usage

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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Core Values," https://framework.scaledagile.com/safe-core-values/
- "Lean-Agile Mindset," https://framework.scaledagile.com/lean-agile-mindset/
- "SAFe Lean-Agile Principles," https://framework.scaledagile.com/safe-lean-agile-principles/
- "AI-Empowered Agility," https://framework.scaledagile.com/ai-empowered-agility

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Lesson 3

Team and Technical Agility

SAFe® Course - Attending this course gives learners access to the SAFe Agilist exam and related preparation materials.

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Lesson Topics

3.1 Forming cross-functional Agile Teams

3.2 Organizing Agile Release Trains around the flow of value

3.3 Built-in quality

3.4 Building a Continuous Delivery Pipeline with DevOps

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3-2

Learning objectives

At the end of this lesson, you should be able to:

- Describe cross-functional Agile Teams
- Explain how Agile Release Trains (ARTs) are organized
- Summarize built-in quality practices
- Explain the need to build and maintain a Continuous Delivery Pipeline with DevOps

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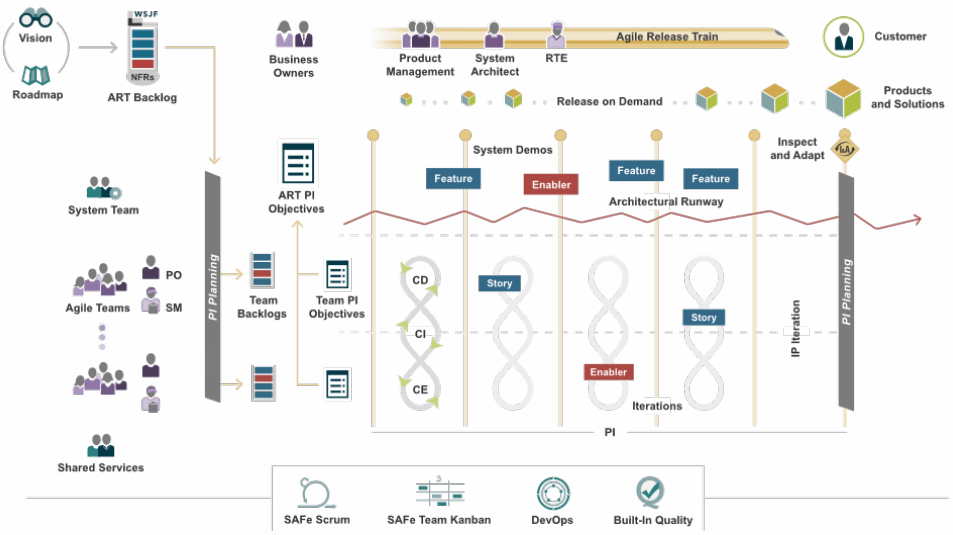
Team and Technical Agility



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3.1 Forming cross-functional Agile Teams

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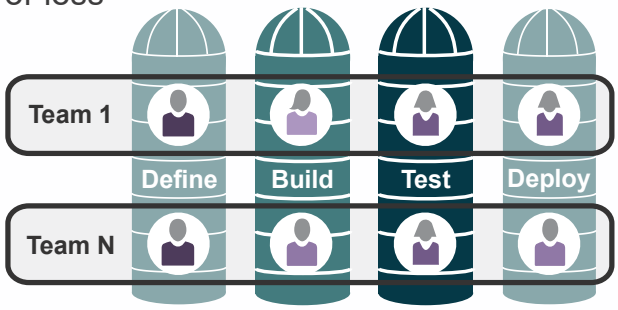
3-5

Build cross-functional Agile Teams

An Agile Team is a cross-functional group of typically ten or fewer individuals with all the skills necessary to define, build, test, and deliver value to their customer.

- Optimized for communication and delivery of value
- Typically include 10 team members or less
- Contain two specialty roles:
  - Scrum Master/Team Coach
  - Product Owner



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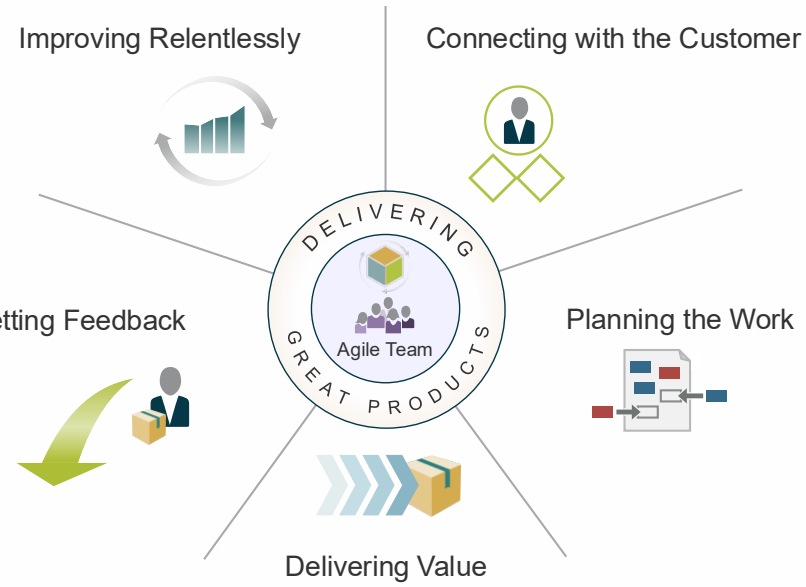
3-6

Agile Team responsibilities



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Agile Teams have two specialty roles

Scrum Master/Team Coach (SM/TC)	Product Owner (PO)
• Facilitates PI Planning • Supports iterations • Improves flow • Builds high-performing teams • Improves ART performance	• Connects with the customer • Contributes to the vision and roadmap • Manages and prioritizes the team backlog • Supports the team in delivering value • Gets and applies feedback

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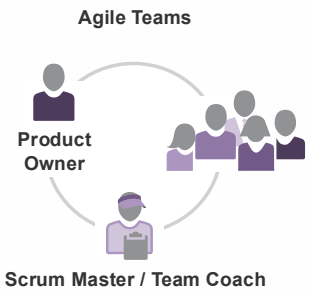
Activity: Form teams for the PI Planning simulation

Duration: 5 min

- **Step 1:** Break into teams. Create a team name in your Team Template.
- **Step 2:** Discuss the Scrum Master/Team Coach responsibilities. Who on your team is best suited to fulfill this role?
- **Step 3:** Discuss the Product Owner responsibilities. Who on your team is best suited to fulfill this role?
- **Step 4:** Make sure the team name and the roles are visible to all other teams in your template.
  - **Note:** In the next lesson, your team will experience Planning Interval (PI) Planning.



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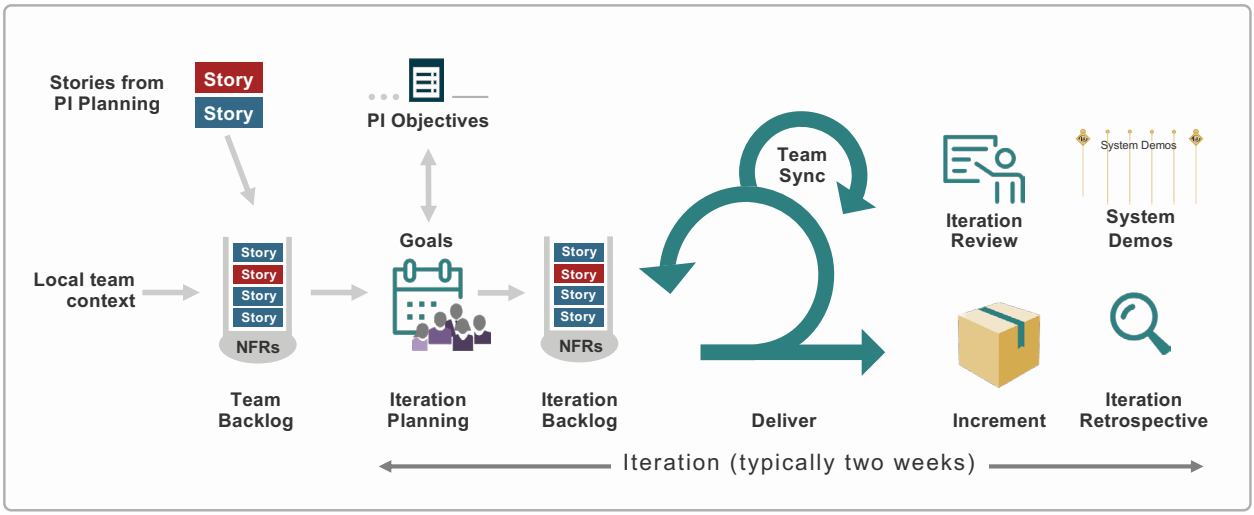
SAFe Scrum overview



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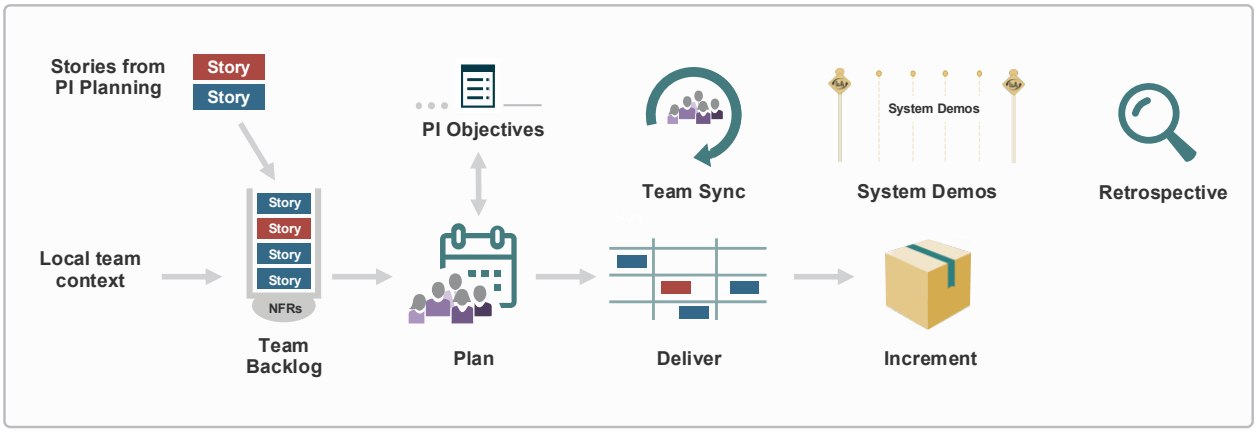
SAFe Team Kanban overview



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All Agile teams focus on value delivery

SAFe Scrum and SAFe Team Kanban promote the delivery of value to customers more efficiently by:

- Working in small batches
- Keeping work-in-process (WIP) under control
- Removing delays
- Incorporating customer feedback to improve the product
- Reviewing and improving practices regularly

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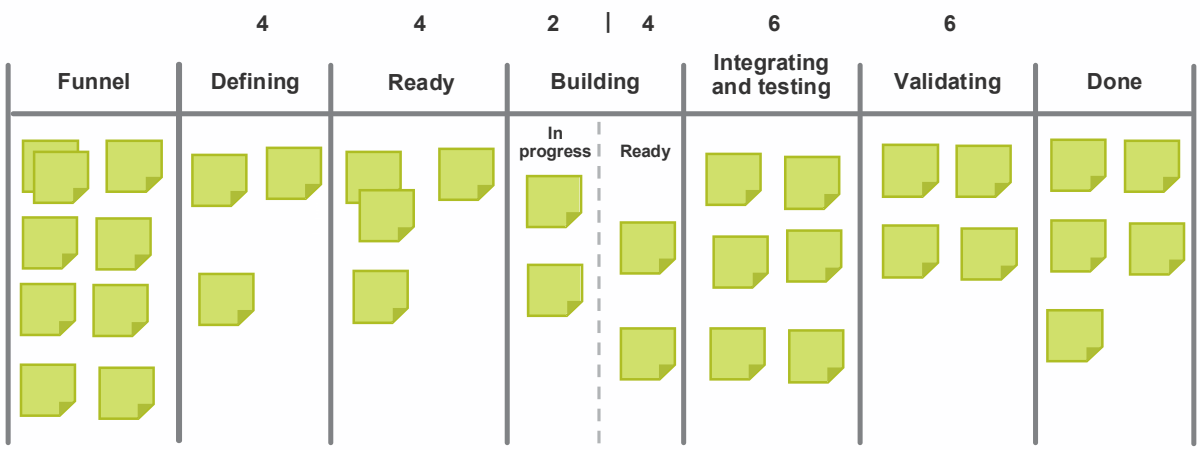
All Agile Teams visualize flow



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Discussion: Empowering Agile Teams with AI

Prepare: 7 min
Share: 3 min

- **Step 1:** Emerging AI capabilities (like agentic AI) can augment how Agile Teams plan, build, test, and learn. With your group, discuss the following:
  - Where could AI meaningfully support your team's work?
  - What guardrails would you want in place?
  - What should remain firmly human-owned?
- **Step 2:** Be prepared to share with the class.

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3.2 Organizing Agile Release Trains around the flow of value

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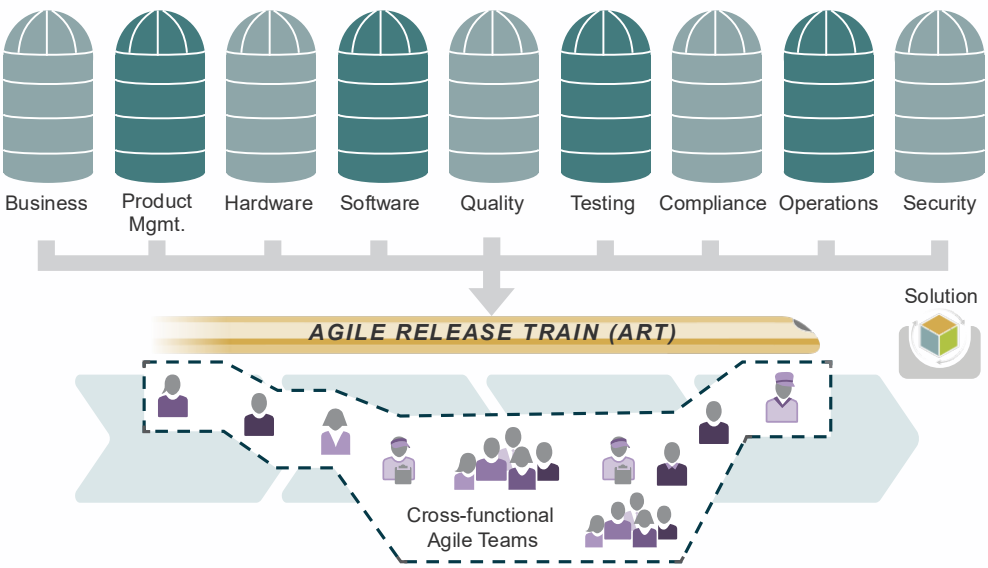
3-15

Agile Release Trains are cross-functional



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3-16

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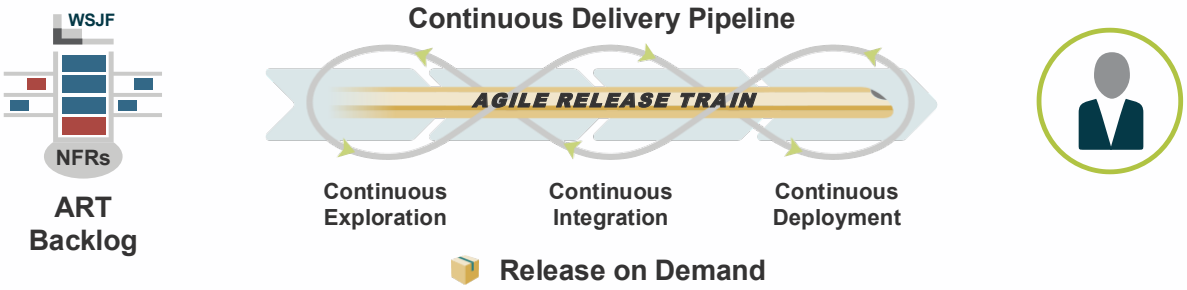
Agile Release Train (ART)

- A virtual organization of 5 – 12 teams (50–125+ individuals)
- Synchronized on a common cadence—a PI
- Aligned to a common mission via a single ART backlog



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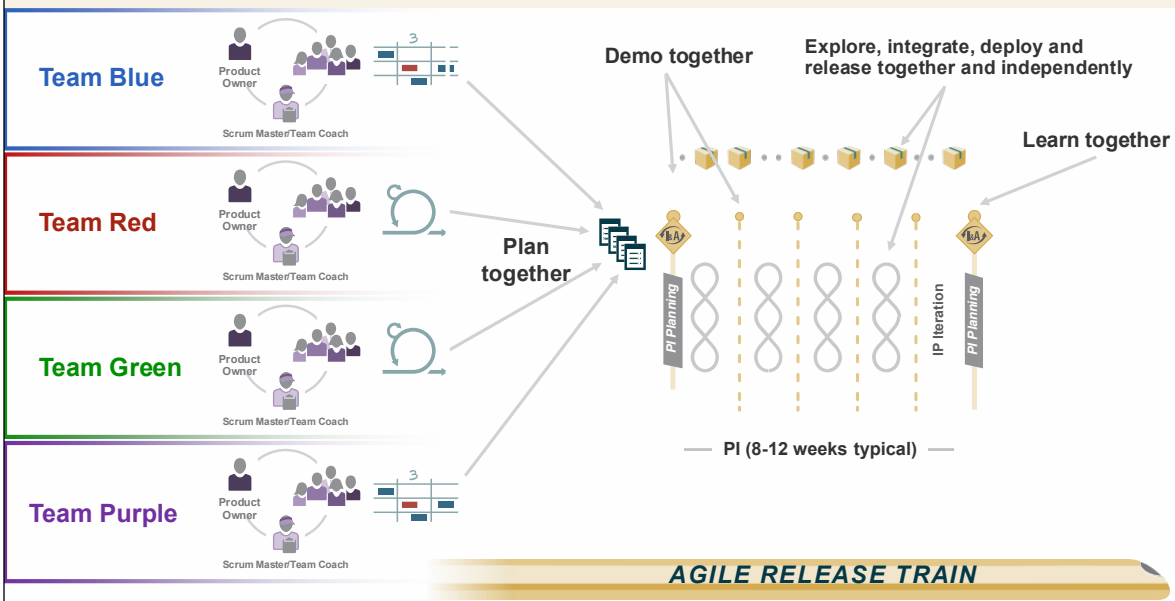
Teams in SAFe are part of an ART



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ART stakeholders

- The **Release Train Engineer** (RTE) is a coach for the ART.
- **Product Management** owns, defines, and prioritizes the ART backlog.
- The **System Architect** provides architectural guidance and technical enablement to the teams on the ART.
- The **system team** provides processes and tools to integrate and evaluate assets early and often.
- **Business Owners** are key stakeholders on the ART.
- **Shared services** represent the specialty roles, people, and services required for the success of an ART but that are not dedicated full-time.

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3-19

3.3 Built-in quality

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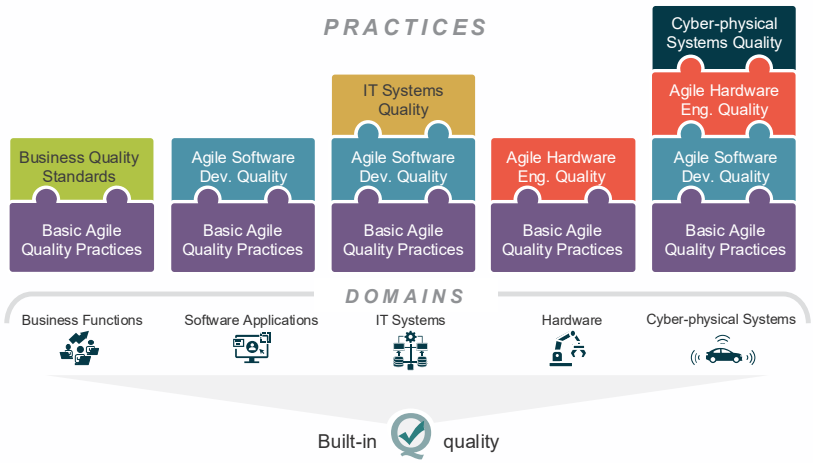
Built-in quality

"You can't scale crappy code (or hardware, or anything else)."

– Dean Leffingwell, Creator of SAFe



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Basic Agile quality practices

Agile quality practices apply to every Agile Team, whether business or technology.

- **Shift learning left:** Reveal problems sooner, take corrective action with minimum impact
- **Pairing and peer review:** Multiple viewpoints enhance work quality and grow knowledge
- **Collective ownership and T-shaped skills:** Reduce bottlenecks and increase flow
- **Artifact standards and definition of done (DoD):** Ensure consistent quality for each work product
- **Workflow automation:** Enable small batches and reduce errors

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AI as a quality accelerator

Quality practices	*What AI can accelerate*
Shift learning left	Earlier signal detection and faster feedback
Pairing and peer review	Expanded review capacity
Collective ownership and T-shaped skills	Lower barriers to contribution
Artifact standards and DoD	More consistent checks
Workflow automation	Fewer manual errors, smaller batches

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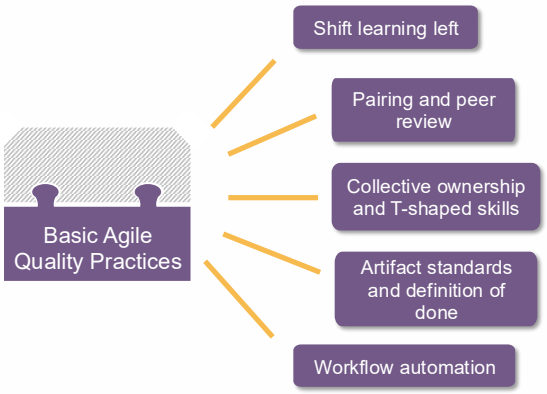
Activity: Reflect on basic Agile quality practices in your context

Prepare: 5 min
Share: 5 min

- **Step 1:** Individually, consider how the basic Agile quality practices apply in your context
- **Step 2**: Identify opportunities to use AI to implement or accelerate the practice
- **Step 3:** Be prepared to share with the class



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Reflect on basic Agile quality practices in your context

Instructions: Reflect on the basic Agile quality practices and consider how they might apply in your context and capture this in the notes section below.

Notes

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3.4 Building a Continuous Delivery Pipeline with DevOps

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Activity: DevOps myth or fact

Prepare: 5 min
Share: 2 min

- **Step 1:** Individually, take the myth or fact quiz in your workbook
- **Step 2:** Check your results with the answer key at the bottom of the page that follows the quiz

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DevOps myth or fact

Instructions: Take this myth or fact quiz individually. Check your results with the answer key at the bottom of the page that follows the quiz.

	Myth	Fact
1. DevOps is just about automation	○	○
2. DevOps is a cultural change	○	○
3. You don't need Lean-Agile to have a successful DevOps implementation	○	○
4. Agile is for development not operations	○	○
5. The deployment pipeline is used to deploy environments as well as solutions	○	○
6. DevOps tries to bridge the gap between new Features and stable solutions	○	○
7. Measurements are an important part of DevOps	○	○
8. Automation of testing reduces the holding cost	○	○
9. DevOps is only for small software companies	○	○
10. Chaos monkey was developed by Netflix	○	○

Notes

Click to reveal answers

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Video: What is DevOps?

Duration: 2 min

https://share.vidyard.com/watch/eM4WBHsac7Cmb2uYqp4Bat

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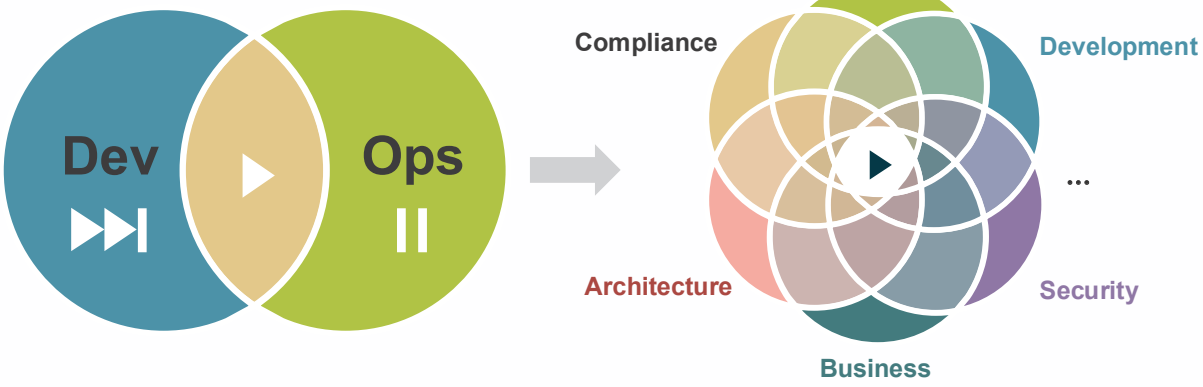
3-27

Maximize speed and stability



Optimized for development speed

Optimized for stability

Working together for speed and stability

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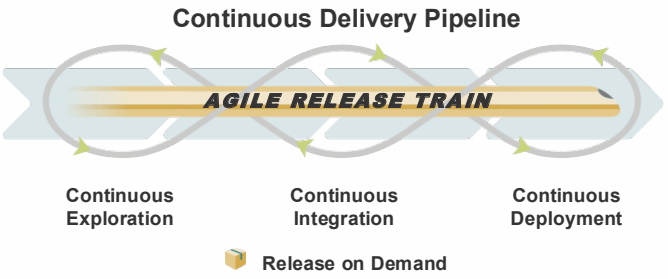
3-28

Building the Continuous Delivery Pipeline with DevOps

- The Continuous Delivery Pipeline (CDP) represents the workflows, activities, and automation needed to deliver new functionality more frequently
- Each ART builds and maintains, or shares, a pipeline
- Organizations map their current pipeline into this new structure and remove delays and improve the efficiency of each step



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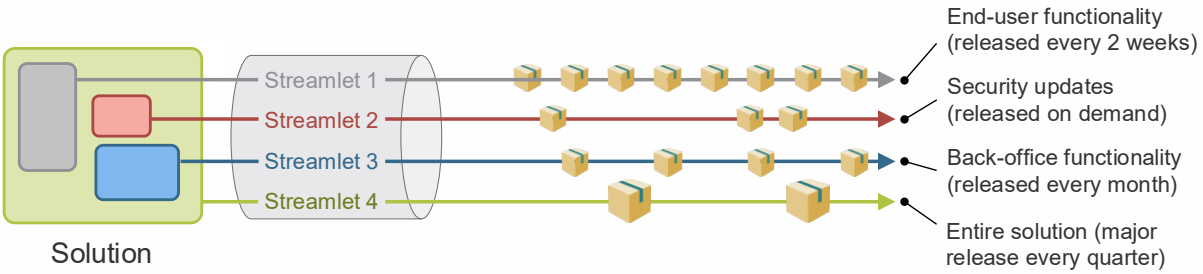
3-29

Decouple release elements from the total solution



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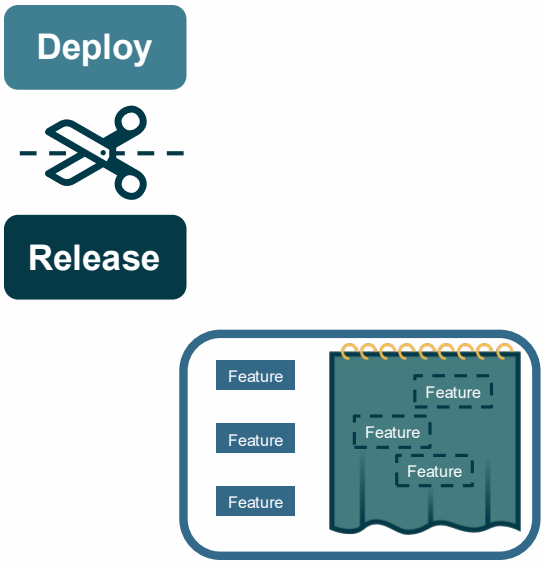
3-30

Separate deploy from release

- Separate deploy to production from release
- Hide all new functionality under feature toggles
- Enables testing background and foreground processes in the actual production environment before exposing new functionality to users
- Timing of the release becomes a business decision



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3-31

Lesson review

In this lesson, you:

- Described cross-functional Agile Teams
- Explained how ARTs are organized
- Described built-in quality practices
- Explained the need to build and maintain a Continuous Delivery Pipeline with DevOps

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3-32

Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson:

- "Team and Technical Agility," https://framework.scaledagile.com/team-and-technical-agility-discipline
- "Built-In Quality," https://framework.scaledagile.com/built-in-quality/
- "Agile Teams," https://framework.scaledagile.com/agile-teams/
- "Agile Release Train," https://framework.scaledagile.com/agile-release-train/
- "DevOps," https://framework.scaledagile.com/devops/
- "Continuous Delivery Pipeline," https://framework.scaledagile.com/continuous-delivery-pipeline/
- "Release on Demand," https://framework.scaledagile.com/release-on-demand/

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Lesson 4

Product Development Flow

SAFe® Course - Attending this course gives learners access to the SAFe Agilist exam and related preparation materials.

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Lesson Topics

4.1 Customer centricity and design thinking

4.2 Prioritizing the ART backlog

4.3 PI Planning

4.4 Executing the PI

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4-2

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Learning objectives

At the end of this lesson, you should be able to:

- Describe the benefits of a customer-centric culture
- Practice applying design thinking
- Craft and refine prompts, and edit AI outputs to support feature writing
- Prioritize the ART backlog with weighted shortest job first (WSJF)
- Participate in a PI Planning simulation

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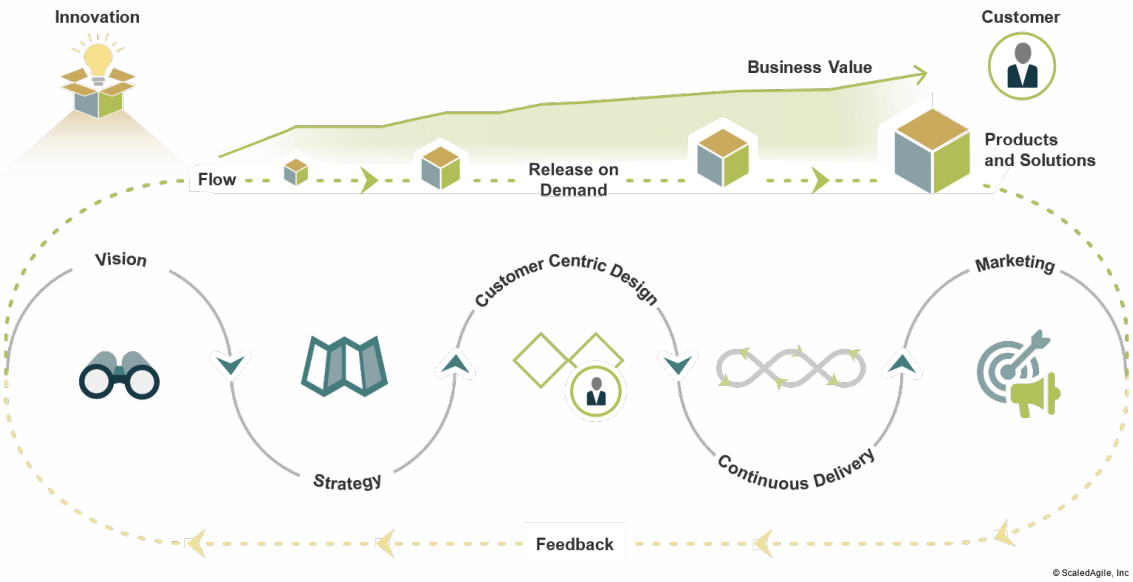
Product Development Flow



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4.1 Customer centricity and design thinking

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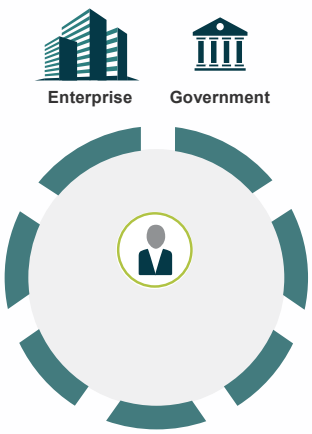
Why customer centricity?

Design whole-product solutions with a deep understanding of customer needs.

Benefits include:

- Greater revenue
- Increased employee engagement
- More satisfied customers
- Alignment to mission
- Better product resiliency and sustainability



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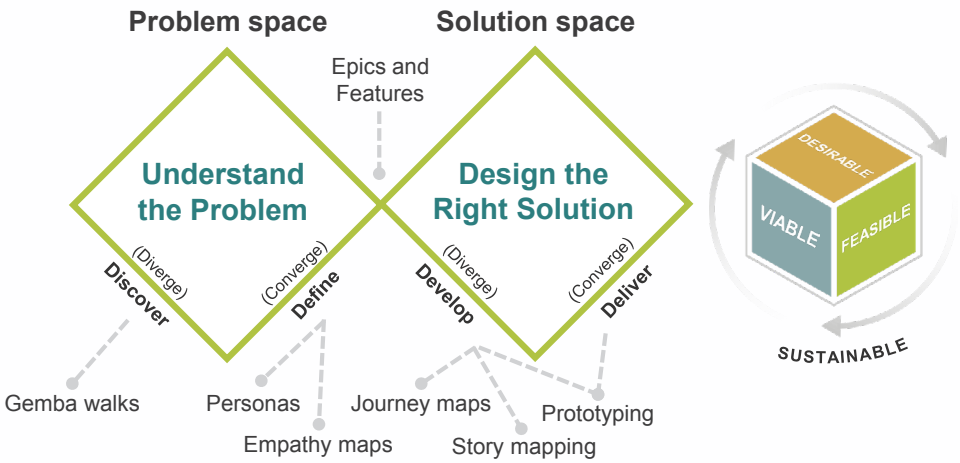
What is design thinking?



Design thinking is a clear and continuous understanding of the target market, customers, the problems they are facing, and the jobs to be done.

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Use personas to understand customers

Personas are characterizations of the people who might use your product. Personas will:

- Convey the problems end users are facing in context, and key triggers for using the product
- Capture rich, concise information that inspires great products without unnecessary details

Cary the Consumer

Age: 36

Location: Reno, Nevada, USA

Time in App: 10 minutes

I'm a working dad with three children ages 3, 6, and 10. I'm also in a band, which means I want to spend as much time as possible with my kids and my band. I need my package delivered on time so that I can maximize time with my family.

- I like technology! I have an iPhone, iPad, and high-speed Wi-Fi
- I am not home on some weekends
- My wife works so she doesn't have much spare time to help
- I'd rather text instead of calling or emailing suppliers

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Use journey maps to design the end-to-end customer experience

	DECIDE *Access options*	LEARN *Clarify goals*	CHOOSE *Choose what to buy*	APPLY *ID & financial check*	PURCHASE *Negotiate sale*	COMPLETE *Celebrate*
Young Couple	Decide to buy a home	Learn about the mortgage / Calculate the budget	Property search / Compare mortgages	Apply for mortgage loan / Assessment	Sign contract & deal	Move
Thinking & Feeling	We loved our neighborhood growing up and want to raise our kids the same way	A real estate agency will save money	Our real estate agent helped us clarify our goals	Just something that will pass quickly / The bank officer will handle everything	We're happy the bank officer kept us updated	
	We've never made a purchase this large before – it is scary!	Visiting bank for consultation = helpful	Selecting real estate agency – comparing & selecting property	Why does the bank need paper copies?	Negotiating a sale this big makes us nervous	We need to clean
		Can we really find a home that we love that we can afford?	Form filing is time consuming & the bank should know our details already	Can't this all be online?	It took a lot longer to close the sale than expected	
Bank		Customer Service Officer • Explains products • Prepares offer • Advises to select real estate agency	Customer Service Officer • Fills in application form & copies ID cards • Provides list with required documents	Credit Department • Performs assessment of applicants • Evaluates property • Prepares approval	Customer Service Officer • Provides updates on loan status	
Technology Opportunities	Online resources for understanding mortgage options	Online advisory, budgeting, financial management, and purchasing options	Automatically gather data from public databases and integrate with social media data	AI-based assessments could improve speed and accuracy	Biometric authentication of electronic signatures	Automated referrals to partners

Customer journey map for a mortgage loan

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4-9

Use empathy maps to identify with customers

- The empathy map is a tool that helps teams develop deep, shared understanding of and empathy for the customer
- Use the empathy map to design better user experiences and Value Streams

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4-10

Learn about your customers with AI-powered research

- Many AI tools offer a deep research mode to explore complex topics
- Deep research uses AI to search, analyze, and synthesize large volumes of information across multiple sources to quickly surface insights, patterns, and summaries
- **Example:** Use deep research prompts to review customer feedback and usage data across multiple channels and highlight recurring themes or issues to address
  - **Reminder:** Deep research outputs must always be reviewed, validated, and acted on by people

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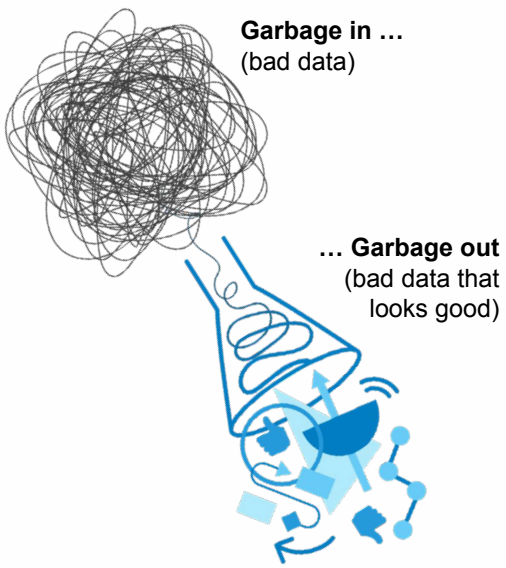
Garbage in: garbage out

- Despite all the headlines about AI breakthroughs, real value still depends on one thing: data.
- Every insight, decision, or prediction generated by AI begins with data. If the data is flawed, incomplete, or biased, then the AI will be too. It may even take bad data and output bad data that looks good.
- The "Garbage In, Garbage Out" principle:
  - Good data = smart, fair, and valuable AI.
  - Bad data = biased, misleading, or even harmful AI.



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Discussion: Exploring customer needs with AI-supported research

Duration: 7 min

- **Step 1:** Individually, review the provided AI-generated customer research report
- **Step 2:** With your group, discuss the following:
  - What assumptions does this research appear to make?
  - What insights would require direct customer validation?
  - How could this research inform (but not replace) customer discovery?
- **Step 3:** Be prepared to share with the class

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4-13

4.2 Prioritizing the ART backlog

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Vision aligns everyone on the product's direction

The product vision is a description of the future state of the product

- How will our product solve our customer's problems?
- What features does it have?
- How will it differentiate us?
- What nonfunctional requirements does it deliver?

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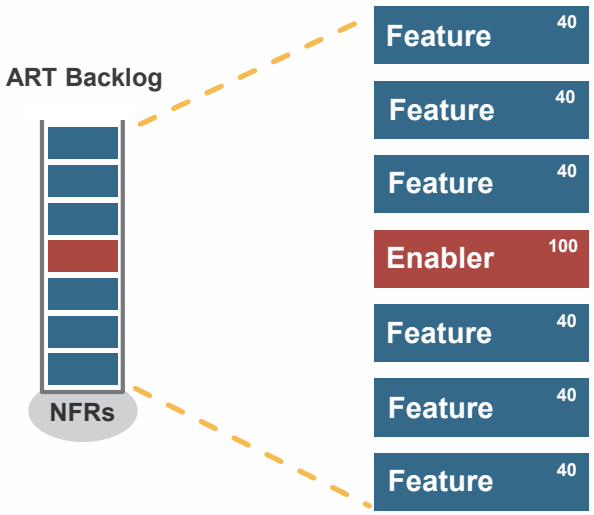
4-15

Features fulfill the product vision

The ART backlog is the holding area for upcoming features that will address user needs and deliver business benefits for a single Agile Release Train (ART).



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4-16

Features represent the work for the Agile Release Train

- The feature benefit hypothesis justifies development costs and provides business perspective for decision-making
- Acceptance criteria are typically defined during ART backlog refinement
- Reflect functional and nonfunctional requirements
- Fits in one PI

Multi-factor authentication

Benefit hypothesis
Enhance user security via both password and a device

Acceptance criteria
1. USB tokens as a first layer
2. Password authentication second layer
3. Multiple tokens on a single device
4. User activity log reflecting both authentication factors

Example feature

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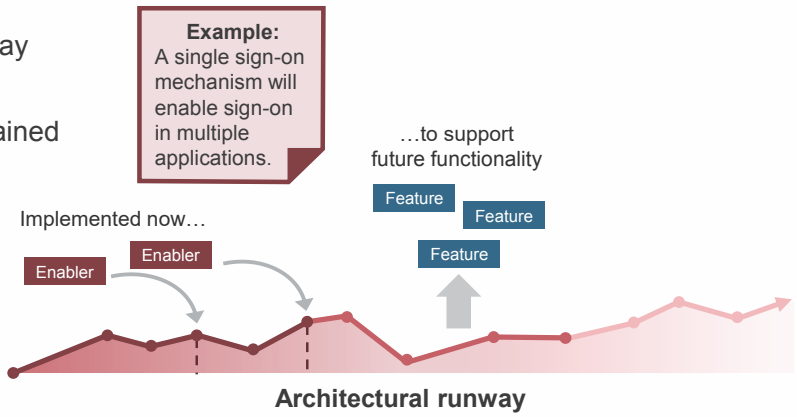
Enablers build the architectural runway

Enablers define the existing code, hardware components, marketing guidelines, and other variables that enable near-term business features.

- Enablers also build up the runway, including exploration, architecture, infrastructure, and compliance
- Features consume the runway
- The architectural runway must be continuously maintained
- Capacity allocation (a percentage of the train's overall capacity in a PI) is used for enablers that extend the runway



Architectural runway

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Activity: Describe a feature

Duration: 5 min

- **Step 1**: Individually, think of a large piece of work from your context and brainstorm one to three potential features and write them in your workbook
- **Step 2:** Add the benefit hypothesis and acceptance criteria and any other important details

Feature:
Multi-factor authentication

Benefit Hypothesis:
Enhance user security via both password and a device

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Using Generative AI to write better features

AI can help clarify intent while ARTs and Agile Teams still own value delivery and acceptance.

- Well-written features improve shared understanding, flow, and feedback
- Generative AI can help clarify audience, intended outcomes, and acceptance criteria
- AI supports drafting and refinement and leaders make final decisions
- SAFe principles guide how AI is used: small batches, fast feedback, and decentralized decisions

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Describe three Features

Instructions: Identify three Features from your context and write them below. If you have time, choose one of the Features and add the benefit hypothesis and acceptance criteria.

Feature: Multi-factor authentication

Benefit Hypothesis: Enhance user security via both password and a device.

Feature:

Benefit Hypothesis:

Feature:

Benefit Hypothesis:

Feature:

Benefit Hypothesis:

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What exactly is a prompt?

- Most generative AI tools require the user to provide a prompt to create the content
- A prompt is the set of instructions you feed an AI system to generate a response
- Effective prompts specify objective, scope, and tone, just as a brief would for a trusted analyst
- Think of it as assigning a task to a high-performing intern: clarity in, clarity out

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4-21

Examples of ways to match your prompt to your purpose:

Purpose	Sample prompt
Feature planning	“Help me brainstorm some features for this work. Start with the user’s end-to-end journey, identify key activities, and suggest candidate user stories for each step.”
Story writing	“Based on this feature description and user persona, draft three to five clear user stories using the user-voice format (“As a… I want… so that).”
Identifying acceptance criteria	“For this user story, suggest clear, testable acceptance criteria that define when the story can be considered complete.”
Brainstorming tasks	“Break this user story into a short list of implementation and validation tasks a team might include in their iteration backlog.”

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4-22

Activity: Prompt, refine, and edit

Prepare: 15 min
Share: 10 min

- **Step 1:** Individually, select one of the features you brainstormed in the previous activity
- **Step 2:** Using a GenAI tool and the AI prompting template on the next slide, write a prompt to help clarify the feature intent, strengthen the benefit hypothesis, and improve acceptance criteria
- **Step 3:** Review the AI-generated output and decide:
  - What to keep
  - What to change
  - What to remove
- **Step 4:** Refine the story and be prepared to share with the class

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4-23

AI prompting template

Basic prompt structure	**Goal:** What are you trying to accomplish?
	Role: What persona or area of expertise should the AI assume in formulating its response?
	Task: What exactly do you want the AI to do?
	Context: Is there background info the AI needs?
	Details: Are there any specific requirements?

NOTE: Intermediate and advanced prompting suggestions are available in the AI Prompting Template to further refine your prompt.

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4-24

Features are implemented by stories

- Stories are small increments of value that can be developed in days and are relatively easy to estimate
- Story user-voice form captures the end user's roles, activities, and goals
- Features fit in one PI for one ART; stories fit in one iteration for one team

Feature

Shipping Method Selection

Benefit hypothesis:
Users can select a shipping method based on cost, delivery speed, and carrier

Enabler story

Research how to calculate the shipping costs.

User story

As a book purchaser **I can** see the price for each shipping method for my current order **so that** I can select a shipping method based on price.

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4-25

Estimate stories with relative story points

- A story point is a number that represents:
  - **Volume** - How much there is
  - **Complexity** - How difficult it is
  - **Knowledge** - What we know
  - **Uncertainty** - What's not known
- Story points are relative, and are not connected to any specific unit of measure

What size is it?

Guidance: Compared to other stories, an eight-point story should take relatively four times longer than a two-point story.

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4-26

Apply estimating poker for fast, relative estimating

- Estimating poker combines expert opinion, analogy, and disaggregation for quick but reliable estimates
- All members participate
- Increases accuracy by including all perspectives
- Builds understanding and creates shared understanding

Steps	
1	Each estimator gets a deck of cards
2	Read a job
3	Estimators privately select cards
4	Cards are turned over
5	Discuss differences
6	Re-estimate

Agile Estimating and Planning by Mike Cohn

?	1	2	3	5	8	13	20	40	100	∞

Warning: Estimation performed by a manager, architect, or select group negates these benefits.

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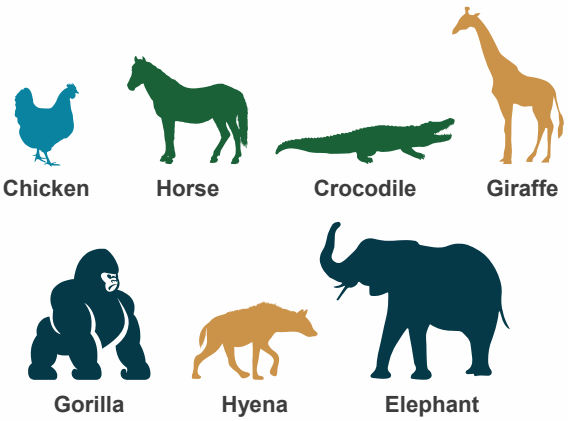
Activity: Relative size estimating

Prepare: 7 min
Share: 3 min

Use estimating poker to relatively estimate the mass of a set of animals.

- **Step 1:** In your groups, identify the smallest animal and mark it as 1
- **Step 2:** Estimate the remaining animals using values 2, 3, 5, 8, 13, 20, 40, and 100



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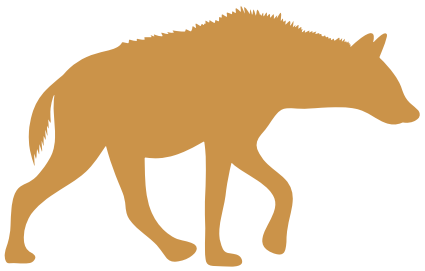
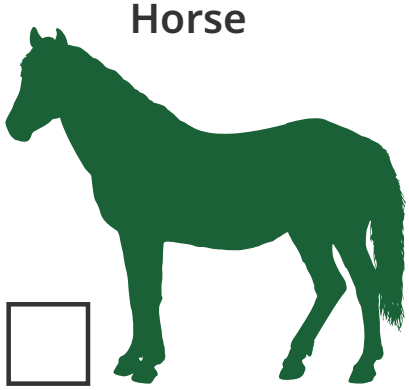
Activity: Relative size estimating

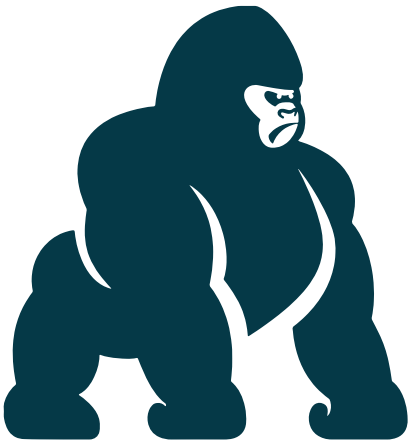
Hyena



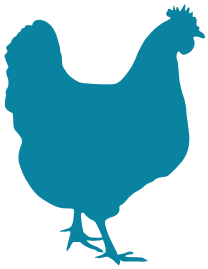
Horse



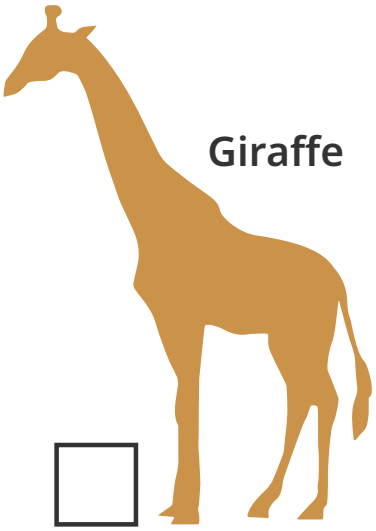
Gorilla



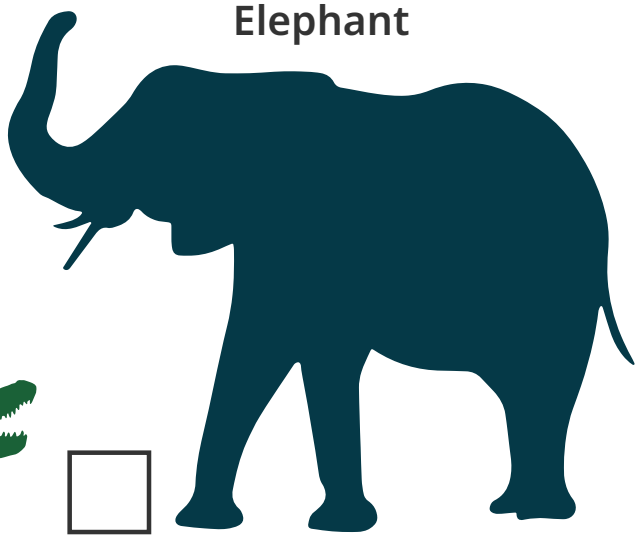
Chicken



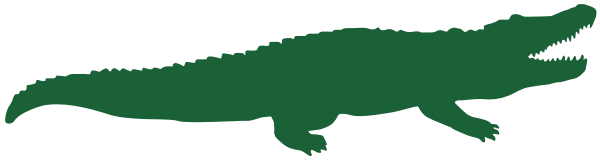
Giraffe



Elephant



Crocodile



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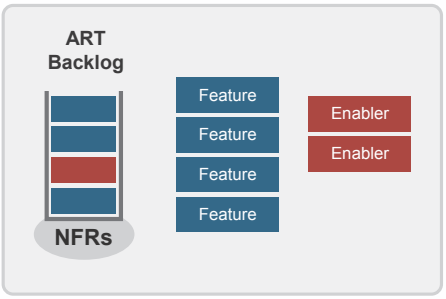
Prioritize features for optimal economic outcomes

Prioritization is the key to improving economic outcomes.

To prioritize features, we need to know two things:

- The cost of delay (CoD) in delivering value
- How long it takes to implement value



"If you only quantify one thing, quantify the cost of delay."

—Donald G. Reinertsen, *The Principles of Product Development Flow*

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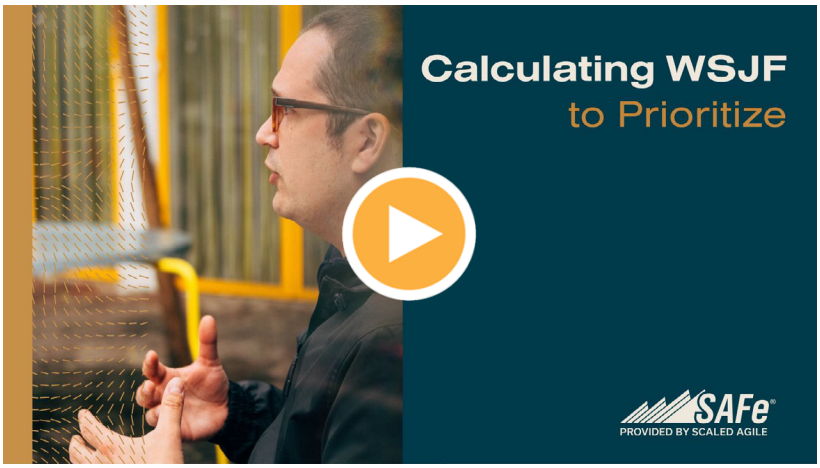
4-29

Optional Video: Calculating WSJF to Prioritize the ART Backlog

Duration: 5 min



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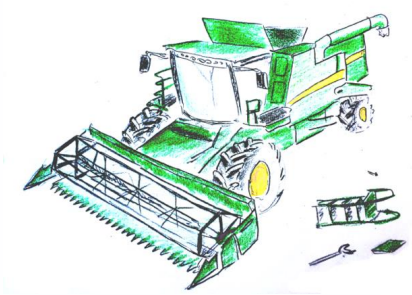
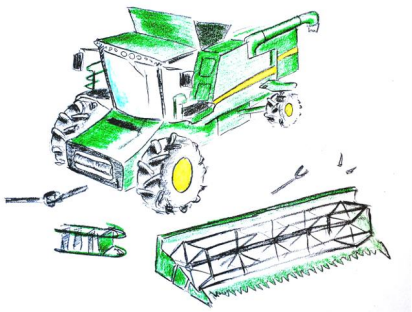
4-30

Example with equal CoD: Which job first?

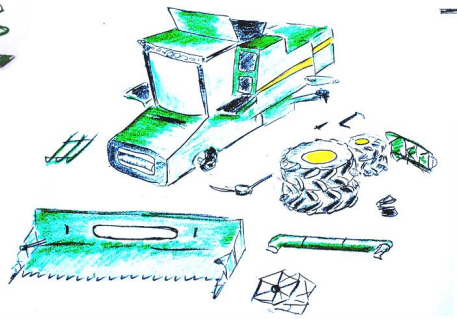
Option A: $$, one day



Option B: $$, three days



Option C: $$, 10 days



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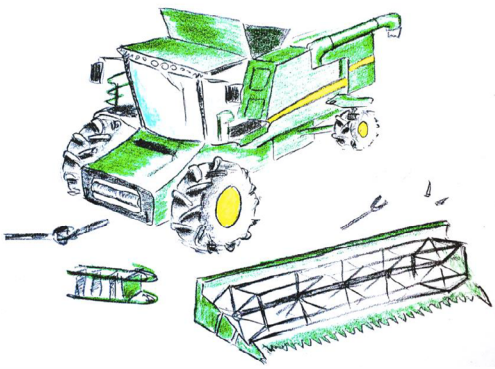
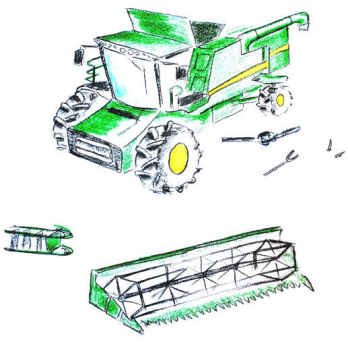
Example with equal duration: Which job first?

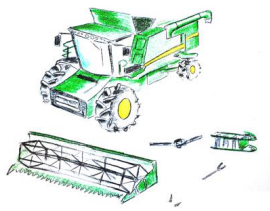
Option A: $$$, three days



Option B: $$, three days



Option C: $, three days



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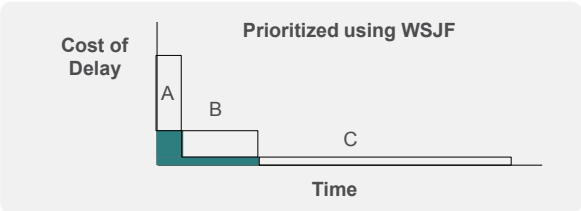
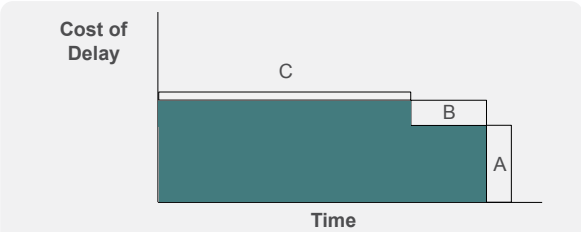
General case: Any CoD and duration

Give preference to jobs with a shorter duration and higher CoD, using WSJF:





$$\text{WSJF} = \frac{\text{Cost of Delay}}{\text{Job Duration}}$$

Feature	Duration	CoD	WSJF
A	1	10	10
B	3	3	1
C	10	1	0.1

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4-33

Components of cost of delay

User-Business Value	Time Criticality	Risk Reduction and/or Opportunity Enablement
What is the relative value to the Customer or business? • Do our users prefer this over that? • What is the revenue impact on our business? • Is there a potential penalty or other negative effects if we delay?	**How does user/business value decay over time?** • Is there a fixed deadline? • Will they wait for us or move to another Solution? • What is the current effect on Customer satisfaction?	**What else does this do for our business?** • Reduce the risk of this or future delivery? • Is there value in the information we will receive? • Enable new business opportunities?

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4-34

Calculate WSJF with relative estimating

In order to calculate WSJF, teams need to estimate cost of delay and duration.

- For duration, use job size as a quick proxy
- Relative estimating is a quick technique to estimate job size and relative value
- WSJF participants: Business Owners, Product Managers, affected teams, and other stakeholders

$$\text{WSJF} = \frac{\text{User-business value} + \text{Time criticality} + \text{Risk reduction and/or opportunity enablement}}{\text{Job size}}$$

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Activity: WSJF prioritization

Prepare: 10 min
Share: 5 min

- **Step 1:** Individually, in your workbook, prioritize three of the features you identified earlier using WSJF
- **Step 2:** Be prepared to share with the class

Feature	User-business value		Time criticality		RR/OE Value	=	CoD	÷	Job size	=	WSJF
Book reviewing	1	+	1	+	1	=	3	÷	3	=	1
Profile management	5	+	5	+	5	=	15	÷	5	=	3
Book rating	3	+	1	+	5	=	9	÷	1	=	9

Scale for each parameter: 1, 2, 3, 5, 8, 13, 20
Note: Do one column at a time. Start by picking the smallest item and giving it a 1. There must be at least one 1 in each column.

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Weighted shortest job first (WSJF) prioritization

Instructions: Prioritize three of the Features you identified earlier using WSJF. Be prepared to share with the class.

$$\text{WSJF} = \frac{\text{User-business value} + \text{Time criticality} + \text{Risk reduction and/or opportunity enablement}}{\text{Job size}}$$

Feature	User-business value	Time criticality	RR | OE value	COD	Job size	WSJF
		+	+	= 0.00 ÷		=
		+	+	= 0.00 ÷		=
		+	+	= 0.00 ÷		=

Scale for each parameter: 1, 2, 3, 5, 8, 13, 20
Note: Do one column at a time. Start by picking the smallest item and giving it a 1. There must be at least one 1 in each column.

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4.3 PI Planning

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Video: SAFe at Travelport: The Power of PI Planning

Duration: 2 min

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What is PI Planning?

Planning Interval (PI) Planning is a cadence-based event that serves as the heartbeat of the Agile Release Train (ART), aligning all teams on the ART to a shared mission and vision.

- Two days every 8–12 weeks (10 weeks is typical)
- Everyone plans together
- Product Management owns feature priorities
- Agile teams own story planning and high-level estimates
- System Architect and User Experience (UX) work as intermediaries for governance, interfaces, and dependencies

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The benefits of PI Planning

- Establishing personal communication across all team members and stakeholders
- Aligning development to business goals with the business context, vision, and Team/ART PI objectives
- Identifying dependencies and fostering cross-team and cross-ART collaboration
- Providing the opportunity for just the right amount of architecture and Lean UX guidance
- Matching demand to capacity; eliminating excess work in process (WIP)
- Fast decision-making

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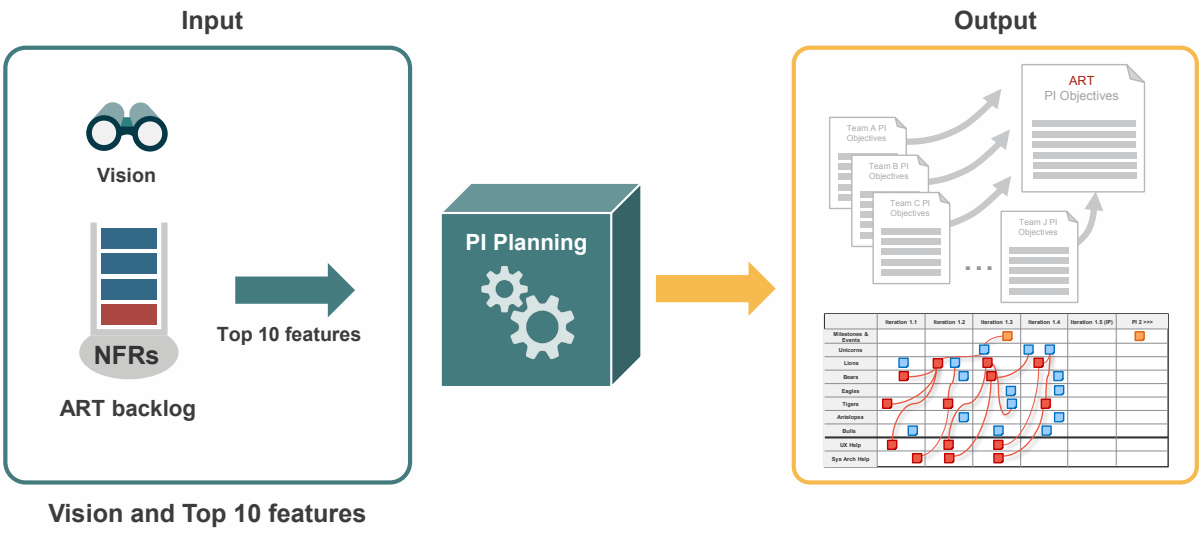
The PI Planning process



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Create alignment with PI objectives

- Objectives are business summaries of what each team intends to deliver in the upcoming PI
- Objectives often directly relate to intended features in the backlog
- Other examples:
  - Aggregation of a set of features
  - A Milestone like a trade show
  - An enabler feature supporting the implementation
- A major refactoring

Objectives for PI 1	BV	AV
1. Show routing calculations between the five most frequent destinations		
2. Navigate autonomously from distribution center to the most frequent destination		
3. Parallel park for a delivery		
4. Return to the distribution center after delivery		
5. Include traffic data in route planning		
6. Recall a delivery that is already in progress		
Uncommitted Objectives		
7. Spike: Reduce GPS signal loss by 25%		
8. Demonstrate real-time rerouting to avoid delays (e.g., accident, construction)		

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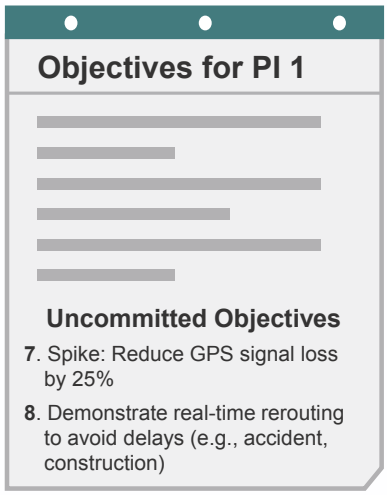
Maintain predictability with uncommitted objectives

Uncommitted objectives help improve the predictability of delivering business value.

- Uncommitted objectives are planned and aren't extra things teams do 'just in case they have time'
- They are not included in the commitment, thereby making the commitment more reliable
- If a team has low confidence in meeting a PI Objective, it should be moved to uncommitted
- If an objective has many unknowns, consider moving it to uncommitted and put in early spikes
- Uncommitted objectives count when calculating load



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Outcomes of the PI Planning simulation

Actively participating in a simulated PI Planning event will reinforce:

Communication
Experience the business benefits of establishing communication across all team members and stakeholders

Drafting Objectives
Experience drafting PI objectives for achieving the Planning Interval and committing to the plan

Estimating Capacity
Experience estimating capacity for the iteration

Managing Risks
Experience managing ART PI risks

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Activity: Identify ART roles

Duration: 3 min

- **Step 1:** Identify ART roles for the simulation in your class
- **Step 2:** Ensure that you have all key roles required for the PI Planning simulation

Note: Your trainer will be the RTE.

Simulation role	Assigned to
Executive	Volunteer
Product Management	Volunteer
System Architect, UX, Development Manager	Volunteer

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Simulation: Why are we here?

Alignment to a common mission

We are here to gain alignment and commitment around a clear set of objectives. We will now review the agenda for the next two days of PI Planning.

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Simulation: Day one agenda

Business context	8:00 – 9:00	• State of the business
Product/Solution Vision	9:00 – 10:30	• Vision and prioritized Features
Architecture Vision and development practices	10:30 – 11:30	• Architecture, common frameworks, etc. • Agile tooling, engineering practices, etc.
Planning context and lunch	11:30 – 1:00	• Facilitator explains the planning process
Team breakouts	1:00 – 4:00	• Teams develop draft plans and identify risks and impediments • Architects and Product Managers circulate
Draft plan review	4:00 – 5:00	• Teams present draft plans, risks, and impediments
Management review and problem solving	5:00 – 6:00	• Adjustments made based on challenges, risks, and impediments

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Simulation: Day two agenda

Planning adjustments	8:00 – 9:00	• Planning adjustments made based on previous day's management meeting
Team breakouts	9:00 – 11:00	• Teams develop final plans and refine risks and impediments • Business Owners circulate and assign business value to team objectives
Final plan review and lunch	11:00 – 1:00	• Teams present final plans, risks, and impediments
ART PI Risks	1:00 – 2:00	• Remaining PI-level risks are discussed and ROAMed
PI confidence vote	2:00 – 2:15	• Team and ART confidence vote
Plan rework if necessary	2:15 – ???	• If necessary, planning continues until commitment is achieved
Planning retrospective and moving forward	After commitment	• Retrospective • Moving forward • Final instructions

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Simulation: Briefings

Executive

Product Manager

System Architect

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Simulation: Planning guidance

Expect this first PI Planning to feel a bit chaotic. Future PI Planning meetings will become more routine.

Product Owners - You have the content authority to make decisions at the **user story** level

Scrum Masters/Team Coaches - Your responsibility is to manage the timebox, the dependencies, and the ambiguities

Agile Team - Your responsibility is to define user stories, plan them into the iteration, and work out interdependencies with other teams

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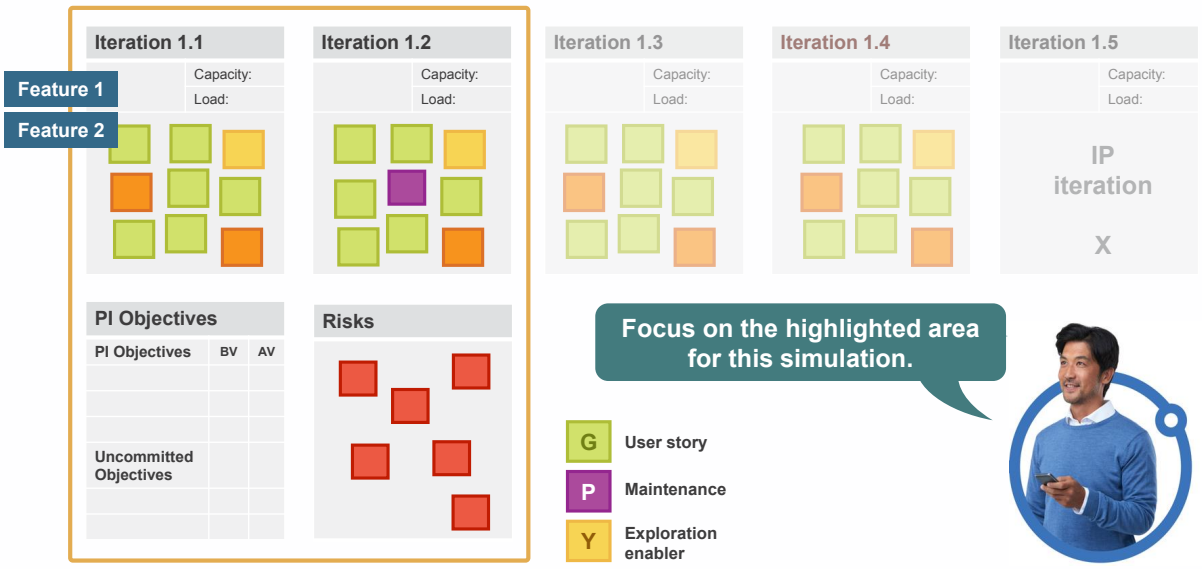
Simulation: Planning requirements



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Activity: Select the feature from Product Management

Duration: 5 min

- **Step 1:** Each team selects a feature from Product Management
- **Step 2:** Ensure this information is visible so it can be referred to during the PI Planning simulation

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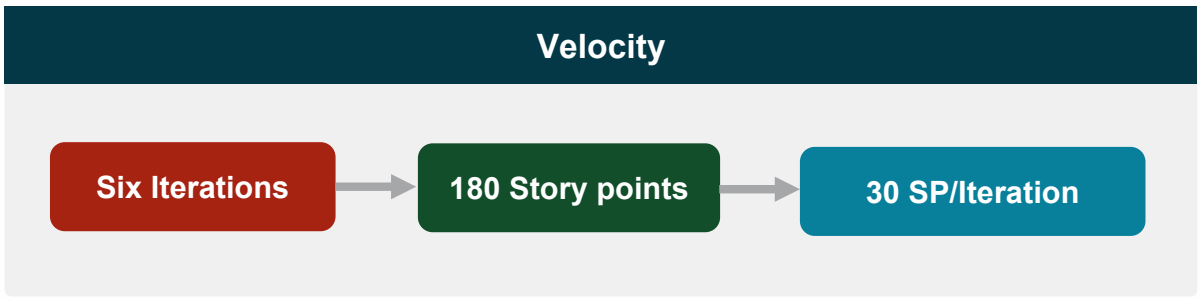
Simulation: Using historical data to calculate velocity

Establish velocity by looking at the average output of the last iterations.



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Simulation: Calculate your initial capacity

- For every full-time Agile Team member contributing to solution development, give the team eight points; adjust this number for part-time team members
- Subtract one point for every team member vacation day and holiday
- Assign one point to a story that would take about a half day to develop and a half day to test and validate
- Estimate every other story relative to that one

Example:
A seven-person team composed of three developers, two testers, one Product Owner, and one Scrum Master/Team Coach
Exclude the Scrum Master/Team Coach, Product Owner, and vacation time from the calculation
Calculated capacity: 5 x 8 points = 40 points per Iteration

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Activity: Team breakout #1

Duration: 50 min

You will be planning a short PI with two iterations with your team.

- **Step 1:** Calculate and enter the capacity for each iteration (two weeks per iteration)
  - The first iteration starts Monday
  - Use your real availability
- **Step 2:** Estimate the stories using story points
- **Step 3:** Load the stories into the iterations
- **Step 4**: Write the PI objectives using clear statements.
- **Step 5:** Identify the uncommitted objectives
- **Step 6:** Identify any ART PI risks and dependencies

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4-55

Activity: Coach Sync

Duration: 5 min

- **Step 1:** The teams observe the Coach Sync, conducted by the RTE
- **Step 2:** Choose a Scrum Master/Team Coach to provide the team's current status and address the questions from the RTE
- **Step 3:** The RTE holds a meet-after following the sync (limited to one or two topics for the simulation)

Note: Coach Sync questions are on the following slide.

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4-56

Activity: Coach Sync

Duration: 5 min

Coach Sync Questions	Team 1
Have you identified the capacity for each Iteration of the PI?	
Have you identified most of the Stories for the first two Iterations and begun estimating?	
Have you begun resolving dependencies with other teams?	
Are you discussing tradeoffs and conflicting priorities with your Business Owners?	
Have you identified any ART PI Risks?	
Will you be ready to start writing PI Objectives in the next 15 minutes?	
Is there anything you need to discuss with other Scrum Masters/Team Coaches? If so, stay for the meet-after.	

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Activity: Draft plan review

Duration: 7 min

- **Step 1:** The teams will present summaries of their first two iterations and one or more draft PI objectives
- **Step 2:** Make sure to include the following:
  - Capacity and load for each iteration
  - Draft PI objectives
  - ART PI risks and impediments

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Management review and problem-solving

At the end of day one, management meets to make adjustments to scope and objectives based on the day's planning.

Common questions:

- What did we just learn?
- Where do we need to adjust vision, scope, team assignments, or anything else?
- Where are the bottlenecks?
- What features must be de-scoped?
- What decisions must we make between now and tomorrow to address these issues?

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Activities during day two

Day One	
Business context	8:00–9:00
Product/Solution Vision	9:00–10:30
Architecture Vision and development practices	10:30–11:30
Planning context and lunch	11:30–1:00
Team breakouts	1:00–4:00
Draft plan review	4:00–5:00
Management review and problem solving	5:00–6:00

Day Two	
Planning adjustments	8:00–9:00
Team breakouts	9:00–11:00
Final plan review and lunch	11:00 –1:00
ART PI Risks	1:00–2:00
PI confidence vote	2:00–2:15
Plan rework if necessary	2:15–???
Planning retrospective and moving forward	After commitment

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Make planning adjustments

- Based on the previous day's management review and problem-solving meeting, adjustments are discussed
- Possible changes:
  - Business priorities
  - Adjustment to vision
  - Changes to scope
  - Realignment of work and teams

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Team breakout #2

Based on new knowledge and a good night's sleep, teams work to create their final plans.

- In the second team breakout, Business Owners circulate and assign business value to PI objectives from low (1) to high (10)
- Teams finalize the PI plan
- Teams also consolidate ART PI risks, impediments, and dependencies
- Uncommitted objectives provide the capacity and guard band needed to increase the reliability of cadence-based delivery

Objectives for PI 1	BV	AV
1. Show routing calculations between the five most frequent destinations	10	
2. Navigate autonomously from distribution center to the most frequent destination	4	
3. Parallel park for a delivery	7	
4. Return to the distribution center after delivery	10	
5. Include traffic data in route planning	7	
6. Recall a delivery that is already in progress	8	
Uncommitted Objectives		
7. Spike: Reduce GPS signal loss by 25%	6	
8. Demonstrate real-time rerouting to avoid delays (e.g., accident, construction)	5	

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Activity: Setting business value

Duration: 7 min

The trainer will demonstrate assigning business value for one team's objectives.

- **Step 1:** Bring one team's draft plans to the Business Owners
- **Step 2:** The Business Owners will set value on a scale of 1–10 for each identified objective
- **Step 3:** Observe the discussion that would take place, illustrating the larger purposes and thought processes around assigning business value

Objectives for PI 1	BV	AV
1. Show routing calculations between the five most frequent destinations	10	
2. Navigate autonomously from distribution center to the most frequent destination	4	
3. Parallel park for a delivery	7	
4. Return to the distribution center after delivery	10	
5. Include traffic data in route planning	7	
6. Recall a delivery that is already in progress	8	
Uncommitted Objectives		
7. Spike: Reduce GPS signal loss by 25%	6	
8. Demonstrate real-time rerouting to avoid delays (e.g., accident, construction)	5	

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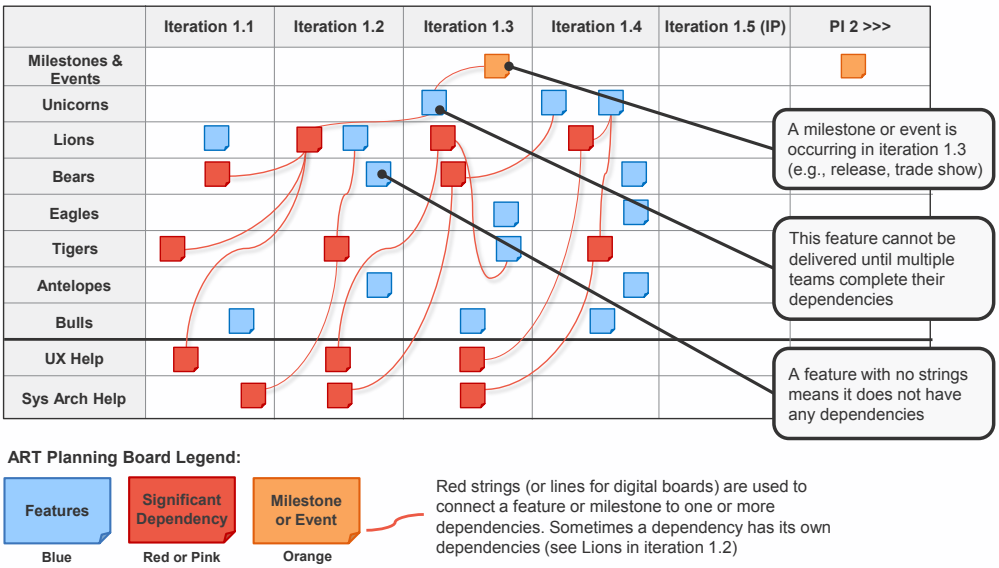
ART planning board: Feature delivery, dependencies, and milestones



ART Planning Board Legend:

Red strings (or lines for digital boards) are used to connect a feature or milestone to one or more dependencies. Sometimes a dependency has its own dependencies (see Lions in iteration 1.2)

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Final plan review

Teams and Business Owners peer-review all final plans.

Final plan review agenda

1. Changes to capacity and load
2. Final PI Objectives with business value
3. ART PI Risks and impediments
4. Q&A session

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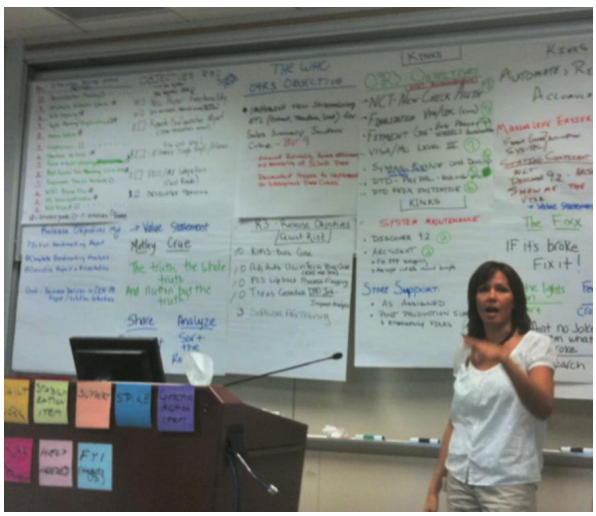
4-65

Building the final plan

- Final plans are reviewed by all teams
- Business Owners are asked whether they accept the plan
- If so, the plan is accepted
- If not, the plan stays in place, and the team continues planning after the review



A team presenting their final plan.
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4-66

Addressing ART PI risks

After all plans have been presented, remaining PI risks and impediments are discussed and categorized.

ROAMing risks:

- **Resolved:** Has been addressed; No longer a concern
- **Owned:** Someone has taken responsibility
- **Accepted:** Nothing more can be done; If risk occurs, release may be compromised
- **Mitigated:** Team has plans to adjust as necessary

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Activity: Manage ART PI risks

Duration: 7 min

Your trainer will demonstrate ROAMing one to two risks for one team.

- **Step 1:** Pick one to two risk examples.
- **Step 2:** Read them in front of all teams and stakeholders.
- **Step 3:** Ask if anyone can own, help mitigate, or resolve the risks. Otherwise, accept the risk as-is.
- **Step 4:** Put each risk into a corresponding quadrant of the ROAM sheet for the ART.

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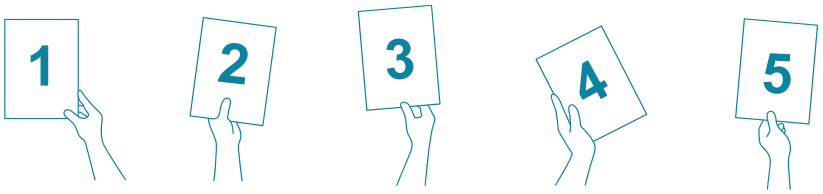
Confidence vote: Teams and ART

Once ART PI risks have been addressed, a confidence vote is taken by each team and the ART.

A commitment with two parts:

1. Teams agree to do everything in their power to meet the agreed-to objectives
2. In the event that objectives are not achievable, teams agree to escalate immediately so that corrective action can be taken



No confidence	Little confidence	Good confidence	High confidence	Very high confidence

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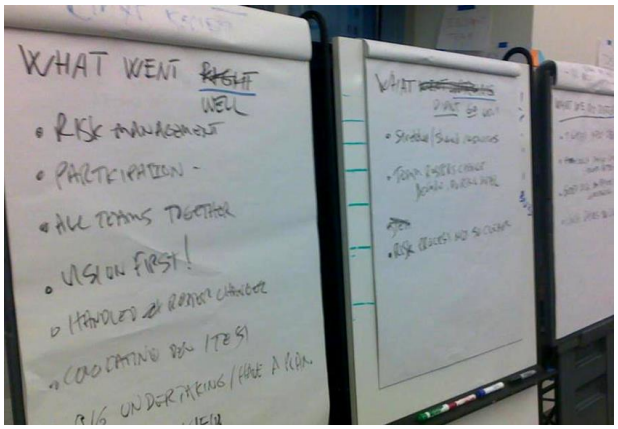
Run a planning meeting retrospective

The PI Planning event will evolve over time. Ending with a retrospective will help continuously improve it.

The PI Planning Event Retrospective

1. What went well
2. What didn't
3. What we can do better next time



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Discussion: AI-empowered PI Planning

Duration: 5 min

As a class, brainstorm ways AI can reduce overhead or friction in PI Planning. Consider:

- Where did planning feel hardest or most time-consuming?
- How could AI reduce effort, improve clarity, or surface risks earlier?
- What should **always** remain a human responsibility?
- What's one AI-enabled improvement you'd consider trying next PI?

Examples:

- Summarizing planning inputs and outputs
- Surfacing risks, dependencies, and timing conflicts
- Capturing decisions and action items in real time

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"Structure is the key to liberation."

—Priya Parker, *The Art of Gathering* (2018)

Photo of Priya Parker. Photo by Jeff Allen, from Parker's website.

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4.4 Executing the PI

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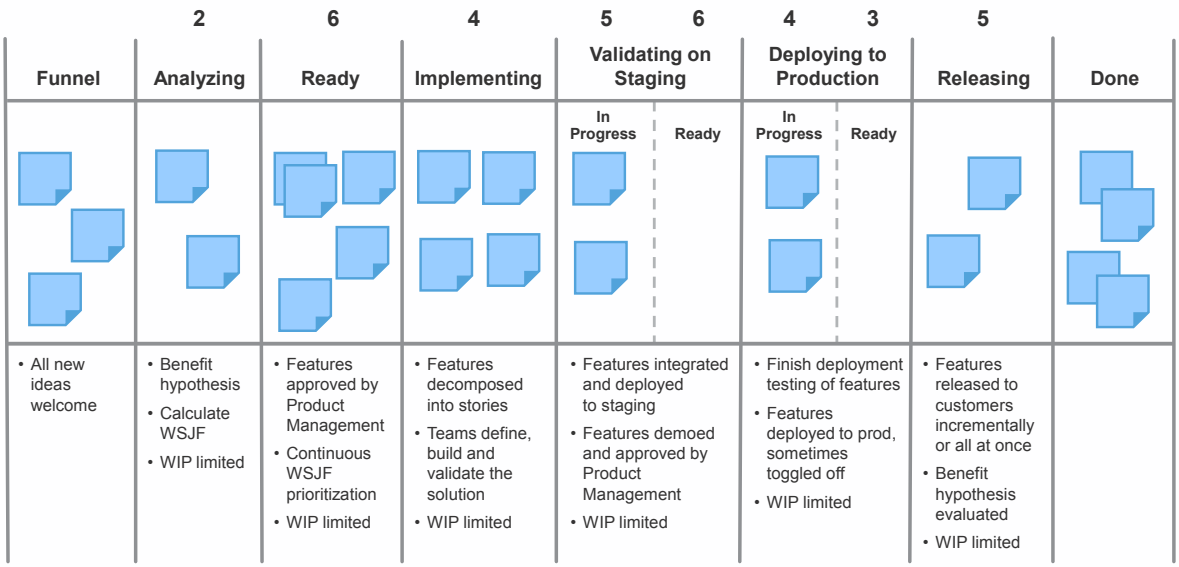
Manage the flow of work with the ART kanban



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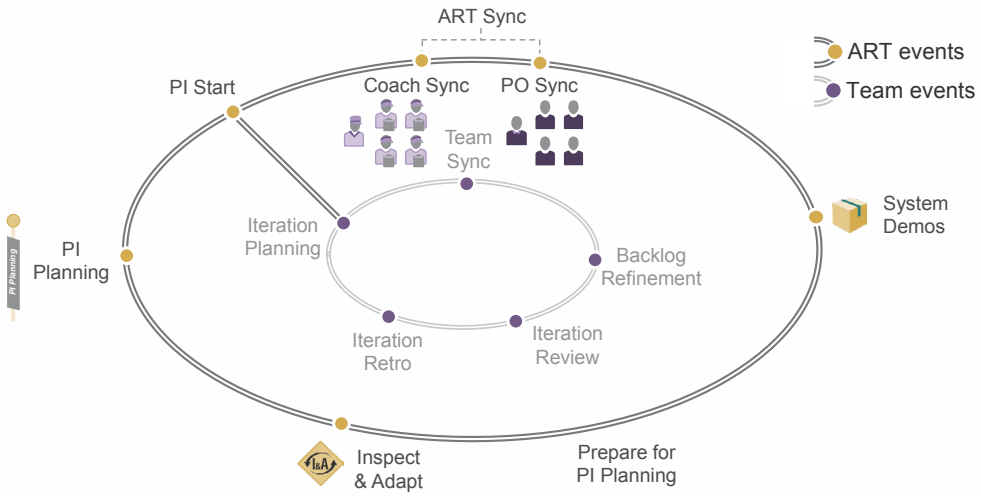
ART events drive the train

Typically facilitated by the RTE, ART events create a closed-loop system to keep the train on the tracks.



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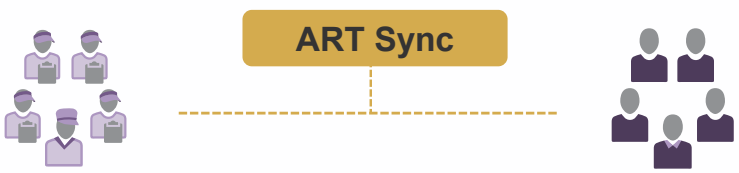
ART Sync is used to coordinate progress



Coach Sync	PO Sync
▶ Visibility into risks, dependencies, progress, and impediments	▶ Visibility into progress, scope, and priority adjustments
▶ Facilitated by RTE	▶ Facilitated by RTE or PM
▶ Participants: Scrum Masters/Team Coaches, other select team members, SMEs if necessary	▶ Participants: PM, POs, other stakeholders, and SMEs as necessary
▶ Weekly or more frequently, 30–60 minutes	▶ Weekly or more frequently, 30–60 minutes
▶ Timeboxed and followed by a meet-after	▶ Timeboxed and followed by a meet-after

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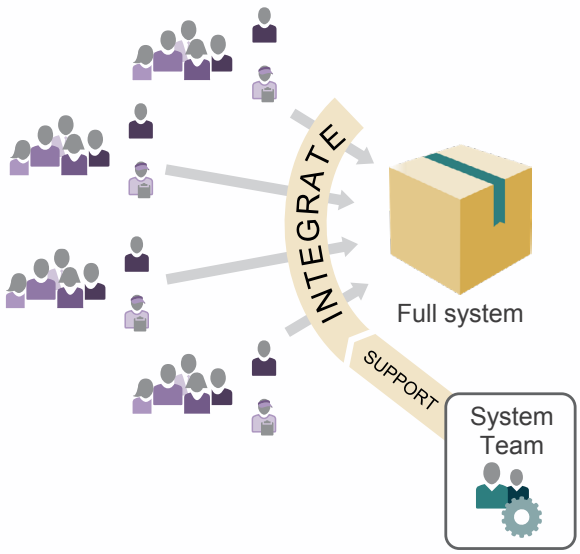
Demo the full system increment every two weeks

- Features are functionally complete or toggled so as not to disrupt demonstrable functionality
- New features work together and with existing functionality
- Happens after the Iteration Review (may lag by as much as one iteration maximum)
- Demo from a staging environment which resembles production as much as possible



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Innovation and Planning (IP) iteration

"Provide sufficient capacity margin to enable cadence."

—Donald G. Reinertsen, *The Principles of Product Development Flow*

Facilitate reliability, PI readiness, planning, and innovation

- **Innovation:** Opportunity for innovation, continuing education, hackathons, and infrastructure improvements
- **Planning:** Provides for cadence-based planning
- Estimating guard band for cadence-based delivery

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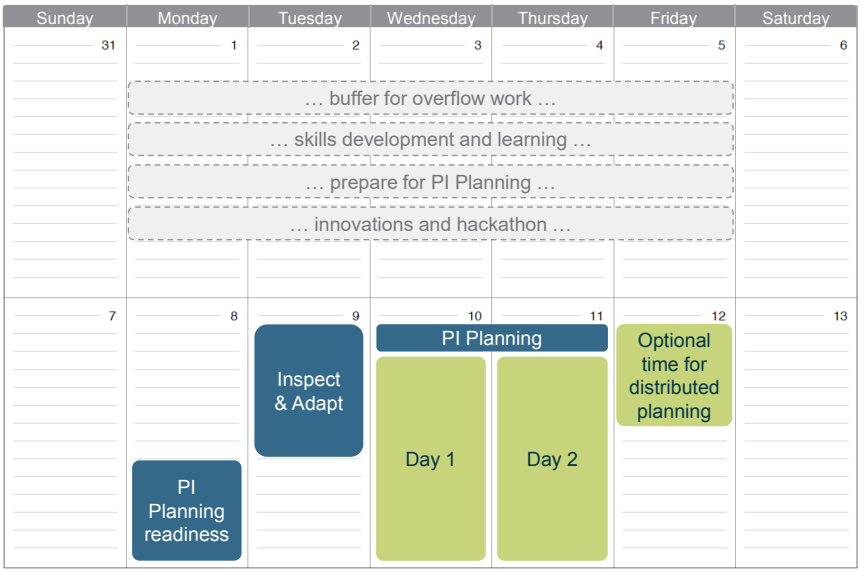
Example IP iteration calendar



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Without the IP iteration...

- Lack of delivery capacity buffer impacts predictability
- Little innovation and 'tyranny of the urgent'
- Technical debt grows uncontrollably
- People burn out
- No time for teams to plan, demo, or improve together

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4-80

Improving results with the Inspect and Adapt event

Three parts of Inspect and Adapt:

1. **The PI System Demo**
2. **Quantitative and Qualitative Measurement**
3. **Problem-Solving Workshop**

- **Timebox:** Three to four hours per PI
- **Attendees:** Teams and stakeholders

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PI System Demo

At the end of the PI, teams demonstrate the current state of the solution to the appropriate stakeholders.

- Often led by Product Management, POs, and the system team
- Attended by Business Owners, ART stakeholders, Product Management, RTE, Scrum Masters/Team Coaches, and teams

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ART performance reporting

Prior to or as part of the PI System Demo, teams review the business value achieved for each of their PI objectives.

- Teams meet with their Business Owners to self-assess the business value they achieved for each objective
- Each team's planned vs actual business value is then rolled up to the ART predictability measure

Objectives for PI 3	Business value Plan	Actual
Structured locations and validation of locations	7	7
Build and demonstrate a proof of concept for context images	8	8
Implement negative triangulation by: tags, companies and people	8	6
Speed up indexing by 50%	10	5
Index 1.2 billion more web pages	10	8
Extract and build URL abstracts	7	7
Uncommitted Objectives		
Fuzzy search by full name	7	0
Improve tag quality to 80% relevance	4	4
Totals	50	45
% Achievement: 90%		

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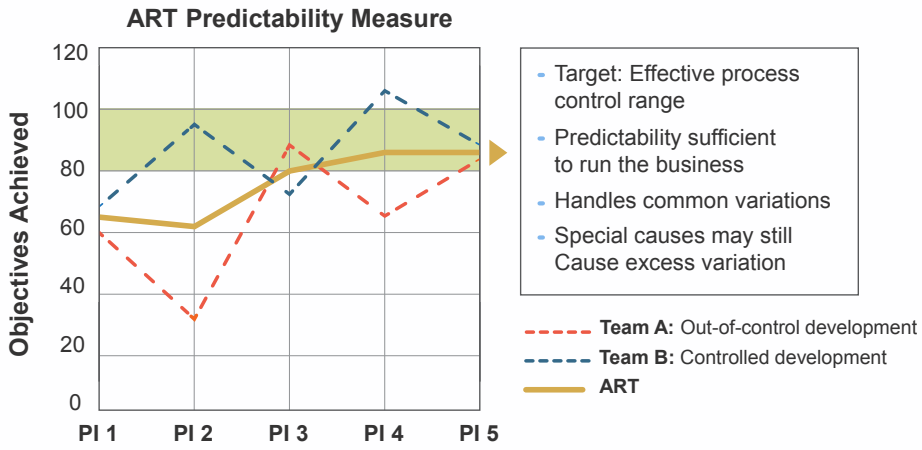
Measure ART predictability

The report compares actual business value achieved against planned business value.



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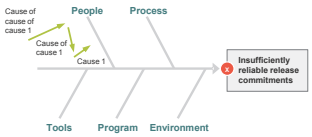
The problem-solving workshop

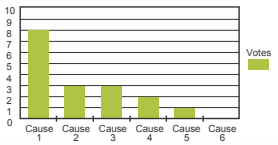
Agree on the problem to solve

X Insufficiently reliable release commitments

Apply root cause analysis and five whys



Identify the biggest root cause using Pareto analysis



Restate the new problem for the biggest root cause

X Insufficient architectural runway

Brainstorm solutions

Identify improvement backlog items

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Lesson review

In this lesson, you:

- Described the benefits of a customer-centric culture
- Practiced applying design thinking
- Crafted and refined prompts, and edit AI outputs to support feature writing
- Prioritized the ART backlog with WSJF
- Participated in a PI Planning simulation

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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Product Development Flow," https://framework.scaledagile.com/product-development-flow-discipline/
- "Design Thinking," https://framework.scaledagile.com/design-thinking/
- "Weighted Shortest Job First," https://framework.scaledagile.com/wsjf/
- "PI Planning," https://framework.scaledagile.com/pi-planning/

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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "ART and Solution Train Backlogs," https://framework.scaledagile.com/art-and-solution-train-backlogs/
- "Story," https://framework.scaledagile.com/Story/
- "PI Objectives," https://framework.scaledagile.com/pi-objectives/
- "Innovation and Planning Iteration," https://framework.scaledagile.com/innovation-and-planning-iteration/
- "Inspect & Adapt," https://framework.scaledagile.com/inspect-and-adapt/

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Lesson 5

Lean Portfolio Management

SAFe® Course - Attending this course gives learners access to the SAFe Agilist exam and related preparation materials.

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Lesson Topics

5.1 Defining a SAFe portfolio

5.2 Connecting the portfolio to the enterprise strategy

5.3 Maintaining the portfolio vision

5.4 Realizing the portfolio vision through epics

5.5 Establishing lean budgets and guardrails

5.6 Establishing portfolio flow

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5-2

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Learning objectives

At the end of this lesson, you should be able to:

- Describe the purpose and elements of a SAFe portfolio
- Construct well-written strategic themes
- Employ the portfolio canvas to describe the current and future state
- Create epic hypothesis statements to inform the vision
- Distinguish traditional and Lean budgeting approaches
- Recognize the steps in the portfolio kanban

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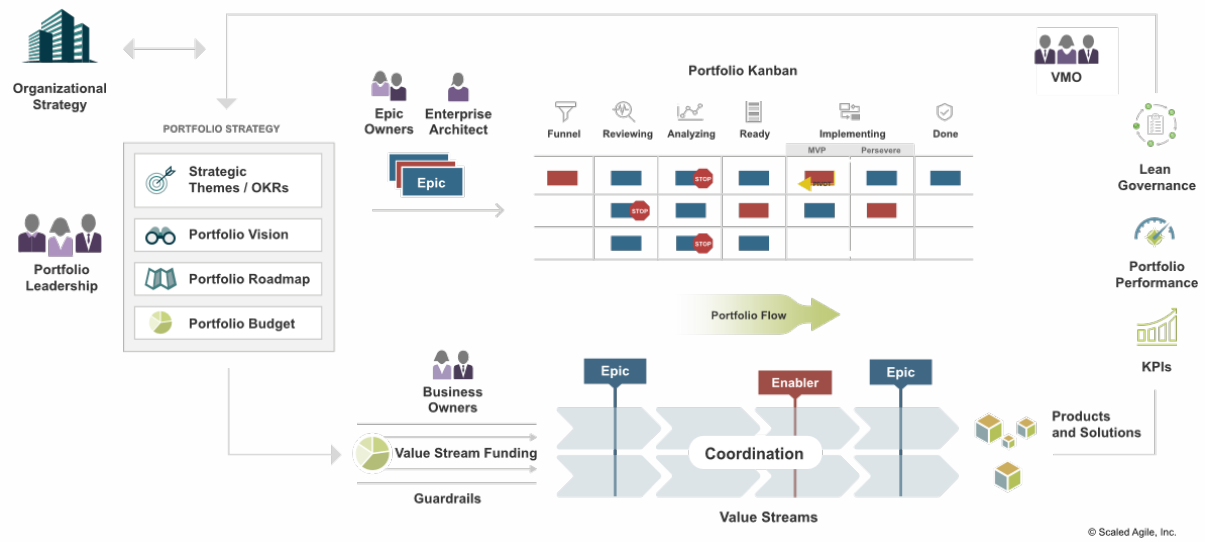
Lean Portfolio Management



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The role of Lean Portfolio Management (LPM)

“ Most strategy dialogues end up with executives talking at cross-purposes because… nobody knows exactly what is meant by vision or strategy, and no two people ever quite agree on which topics belong where. That is why, when you ask members of an executive team to describe and explain the corporate strategy, you so frequently get wildly different answers. We just don’t have good business discipline for converging on issues this abstract.”

—Geoffrey A. Moore, *Escape Velocity*

Portrait of Geoffrey A. Moore. Reproduced with permission of Nancy Warner.

5-5

5.1 Defining a SAFe portfolio

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5-6

What is a SAFe portfolio?

A SAFe portfolio is a collection of development value streams.

- Each value stream builds, supports, and maintains solutions
- Solutions are delivered to the customer, whether internal or external to the enterprise

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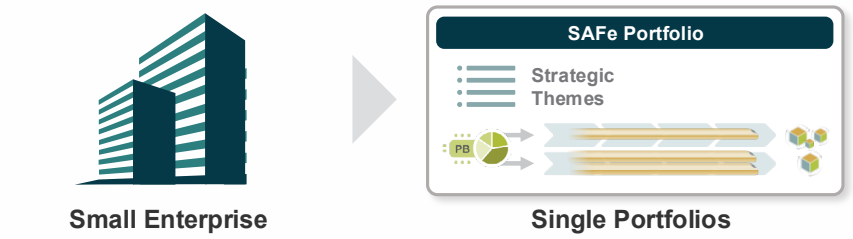
5-7

An enterprise may have a single portfolio or multiple portfolios



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5-8

Define the portfolio with the portfolio canvas

- The portfolio canvas is a template for identifying a specific SAFe portfolio
- It defines the domain of the portfolio and other key elements

Portfolio Canvas Portfolio Name: Date: Version:

Value Propositions

Value Streams	Solutions	Customer Segments	Channels	Customer Relationships	Budget	KPIs / Revenue
What are the value streams in this portfolio? (Note: Create a row for each value stream)	What Solutions does each provide?	What Customers does each value stream serve?	What channels does each value stream use to reach its Customers?	What type of Customer relationship does each value stream maintain?	What is the budget for each value stream?	What measures are used to evaluate the performance of each value stream?

Key Partners

Who are our key partners?
Who are our key suppliers?
Which key resources are we acquiring from partners?
Which key activities do partners perform?

Key Activities

What key activities do our value propositions require?
- Strategic Themes and Lean Budgets
- Market rhythms
- Portfolio sync
- PI Planning (pre/post)
- System/Sol. Demo
- Inspect & Adapt

Key Resources

What key resources do our value propositions require?
- Epic Owners
- Enterprise Architects
- LPM authorities
- VMO, LACE
- Shared services

Cost Structure

What is the total portfolio budget?
What are the most important costs inherent in the portfolio?
Which key resources are most expensive?
Which key activities are the most expensive?

Revenue Streams

What is the revenue for value streams that are monetized?
- For what value are customers willing to pay?
- For what do they currently pay?
- How are they currently paying?
- How much does each revenue stream contribute to overall revenue?

What is the value provided by non-monetized (pure development) value streams?

The Portfolio Canvas is adapted from The Business Model Canvas by Strategyzer (https://www.strategyzer.com/canvas/business-model-canvas).

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5-9

5.2 Connecting the portfolio to the enterprise strategy

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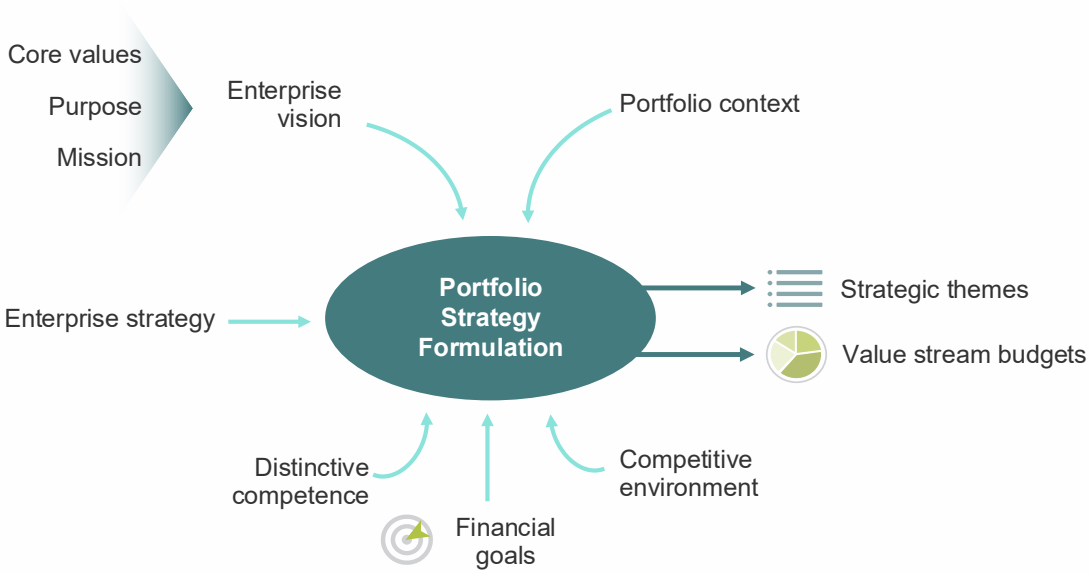
5-10

Elements of portfolio strategy formulation



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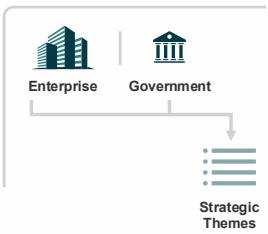
Connect the portfolio to the enterprise with strategic themes



Strategic Themes are differentiating business objectives that:

- Are a collaboration between LPM and the larger enterprise
- Drive the future state of a portfolio
- Connect the portfolio to the enterprise strategy
- Provide context for the portfolio vision and Lean budgeting

Good examples of strategic themes	Bad examples of strategic themes
Expand autonomous delivery into retail	Increase shareholder wealth
Expand autonomous delivery into healthcare	Maximize corporate wealth

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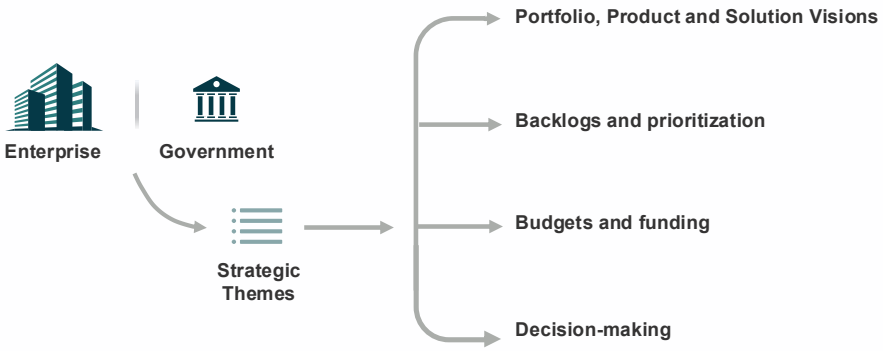
Influence of strategic themes

Strategic Themes influence portfolio strategy and provide business context for portfolio decision making.



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Activity: Identify strategic themes

Prepare: 5 min
Share: 3 min

- **Step 1:** Individually, in your workbook, identify three strategic themes that help define the strategy of your portfolio in the upcoming year
  - Are these differentiators for your business, or business as usual?
- **Step 2:** Be prepared to share with the class

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Identify Strategic Themes

Instructions: Identify three Strategic Themes that help define the strategy of your portfolio in the upcoming year. Are these differentiators for your business, or business as usual?

Strategic Theme #1

Strategic Theme #1

Strategic Theme #1

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5.3 Maintaining the portfolio vision

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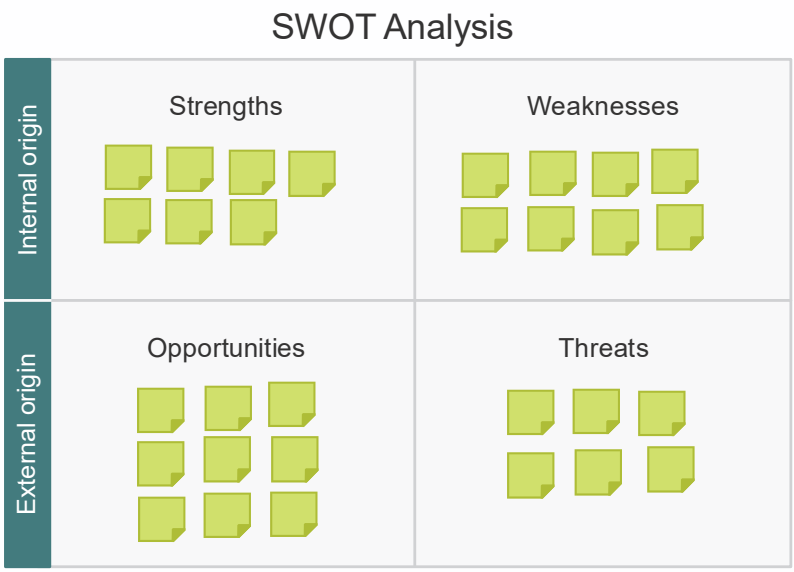
Identify opportunities for the portfolio's future state with SWOT

- Establishes an understanding of the portfolio's strengths and weaknesses
- Helps identify the most significant opportunities and potential threats



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5-16

TOWS strategic options matrix

- The key difference between the SWOT and TOWS analyses are the outcomes that they create
- A TOWS analysis is used primarily for identifying strategic options to create a better future state
- A SWOT analysis is a great way to uncover the current situation of your value stream, product, or portfolio

	External Opportunities (O) 1. 2. 3. 4.	External Threats (T) 1. 2. 3. 4.
Internal Strength (S) 1. 2. 3. 4.	SO How can your strengths be used to exploit and maximize opportunities?	ST How can you apply your strengths to overcome present and potential threats?
Internal Weaknesses (W) 1. 2. 3. 4.	WO How can your opportunities be leveraged to overcome weaknesses?	WT How can you minimize weaknesses and avoid threats?

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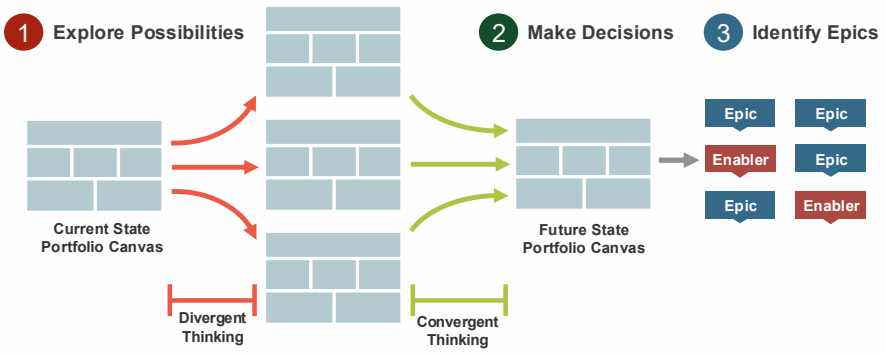
5-17

Envision the future state

- The portfolio canvas captures the current state
- Use SWOT and TOWS to brainstorm potential future states
- Evaluate your options and select a future state
- Identify the epics that will get you to this future state



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5-18

Express the future state as a vision

A long view:

- How will our portfolio of future solutions solve the larger customer problems?
- How will these solutions differentiate us?
- What is the future context within which our solutions will operate?
- What is our current business context, and how must we evolve to meet this future state?

Vision:
A postcard from the future

- Aspirational, yet realistic and achievable
- Motivational enough to engage others on the journey

Result: Everyone starts thinking about how to apply their strengths in order to get there.

Switch: How to Change Things When Change Is Hard, Heath and Heath, Broadway Books, 2010

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5.4 Realizing the portfolio vision through epics

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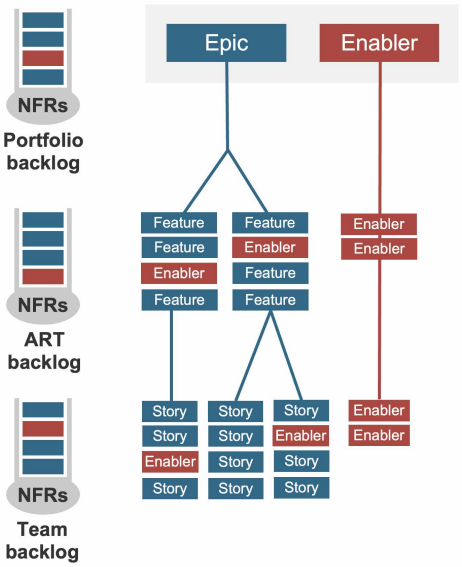
What is a portfolio epic?

- An epic is a significant solution development initiative. There are two types:
  - Business epics directly deliver business value.
  - Enabler epics support the architectural runway and future business functionality.
- Portfolio epics are typically cross-cutting, typically spanning multiple value streams and PIs.



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Epics are initially described with an epic hypothesis statement

Epics are described with four major fields:

- **Epic Hypothesis Statement:** Describes the epic in detailed terms, including the "for-who-the…" portion
- **Business Outcomes:** States the quantitative or qualitative benefits that the business can anticipate if the hypothesis is proven to be correct
- **Leading Indicators:** Describe the early measures that will help predict the business outcomes
- **Nonfunctional Requirements (NFRs):** Identify any NFRs associated with the epic

Epic Hypothesis Statement	
Funnel Entry Date:	[The date that the epic entered the funnel.]
Epic Name:	[A short name for the epic.]
Epic Owner:	[Name of the Epic Owner.]
Epic Description:	[An elevator pitch (value statement) that describes the epic in a clear and concise way.] **For** [customers] **who** [do something] **the** [solution] **Is a** [something – the 'how'] **that** [provides this value] **unlike** [competitor, current solution or non-existing solution] **our solution** [does something better – the 'why']
Business Outcomes:	[The measurable benefits that the business can anticipate if the epic hypothesis is proven to be correct.]
Leading Indicators:	[The early measures that will help predict the business outcome hypothesis. For more on this topic, see the Innovation Accounting advanced topic article.]
Nonfunctional Requirements (NFRs):	[Nonfunctional requirements (NFRs) associated with the epic.]

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Challenge and refine with AI

AI can help Epic Owners and portfolio leaders improve epic hypothesis statement quality by:

- Checking clarity of the for-who / what / why hypothesis
- Identify missing or weak business outcomes
- Surface unclear or lagging leading indicators
- Highlight potential NFR gaps or assumptions
- Expose optimism bias or vague success criteria
- Identify implicit assumptions in the hypothesis

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Activity: Epic writing

Duration: 10 min

- **Step 1:** Individually, in your workbook, identify an epic from your context
- **Step 2:** Write the epic hypothesis statement

Epic Hypothesis Statement	
Funnel Entry Date:	[The date that the epic entered the funnel.]
Epic Name:	[A short name for the epic.]
Epic Owner:	[Name of the Epic Owner.]
Epic Description:	[An elevator pitch (value statement) that describes the epic in a clear and concise way.] **For** [customers] **who** [do something] **the** [solution] **Is a** [something – the 'how'] **that** [provides this value] **Unlike** [competitor, current solution or non-existing solution] **our solution** [does something better – the 'why']
Business Outcomes:	[The measurable benefits that the business can anticipate if the epic hypothesis is proven to be correct.]
Leading Indicators:	[The early measures that will help predict the business outcome hypothesis. For more on this topic, see the Innovation Accounting advanced topic article.]
Nonfunctional Requirements (NFRs):	[Nonfunctional requirements (NFRs) associated with the epic.]

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Activity: Use AI to refine the epic hypothesis statement

Prepare: 10 min
Share: 5 min

- **Step 1:** With an LLM tool, use the following prompt to test your statement:
  - "Evaluate this epic hypothesis statement. Identify the assumption being tested, the leading indicators, and any missing success measures.".
- **Step 2:** Evaluate the feedback and decide with suggestions to incorporate.
- **Step 3:** Be prepared to share with the class.

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5.5 Establishing lean budgets and guardrails

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Epic writing

Instructions: Identify an Epic from your context. Write the Epic hypothesis statement.

Epic: Develop next-generation van hardware to capture additional data on van performance

Epic: Maintenance programs tailored for each van based on sensor data

Epic: Automated electronic inspections and tracking

Epic Hypothesis Statement	
Funnel Entry Date:	<The date that the epic entered the funnel.>
Epic Name:	<A short name for the epic.>
Epic Owner:	<The name of the epic owner.>
Epic Description:	<An elevator pitch (value statement) that describes the epic in a clear and concise way.> For <customers> who <do something> the <solution> is a <something – the 'how'> that <provides this value> unlike <competitor, current solution or non-existing solution> our solution <does something better — the 'why'>
Business Outcomes:	<The measurable benefits that the business can anticipate if the epic hypothesis is proven to be correct.>
Leading Indicators:	<The early measures that will help predict the business outcome hypothesis. For more on this topic, see the Innovation Accounting advanced topic article.>
Nonfunctional Requirements (NFRs):	<Nonfunctional requirements (NFRs) associated with the epic.>

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Epic writing

Epic Hypothesis Statement	
Funnel Entry Date:	
Epic Name:	
Epic Owner:	
Epic Description:	
Business Outcomes:	
Leading Indicators:	
Nonfunctional Requirements (NFRs):	

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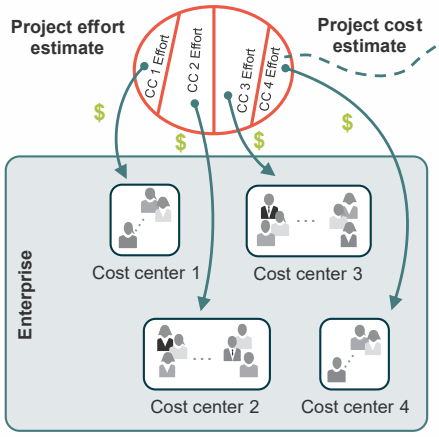
Problem: Cost-center budgeting

Traditional project-based, cost-center budgeting creates overhead and friction, and lowers velocity.



Project

A project requires collaboration of cost centers, assignment of people, budget, and schedule.
It takes multiple budgets to build a single project budget.

Result:

- Slow, complex budgeting process
- Leads to utilization-based planning and execution
- Low throughput
- Moves people to the work

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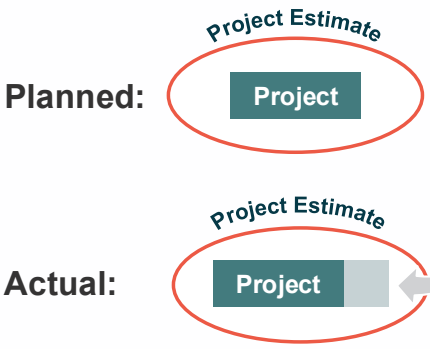
Project overruns cause re-budgeting and increase costs of delay



What/who do we blame?

- Technology challenge?
- Change in scope?
- The team?

Does it matter?

Result:

- Wait for new budget approval; increase cost of delay (CoD)
- Costly variance analysis; blame game; threatens transparency
- Resource scramble reassignments

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Solution: fund value streams, not projects

Funding value streams provides for full control of spend, with:

- No costly and delay-inducing project cost variance analyses
- No resource reassignments
- No blame game for project overruns

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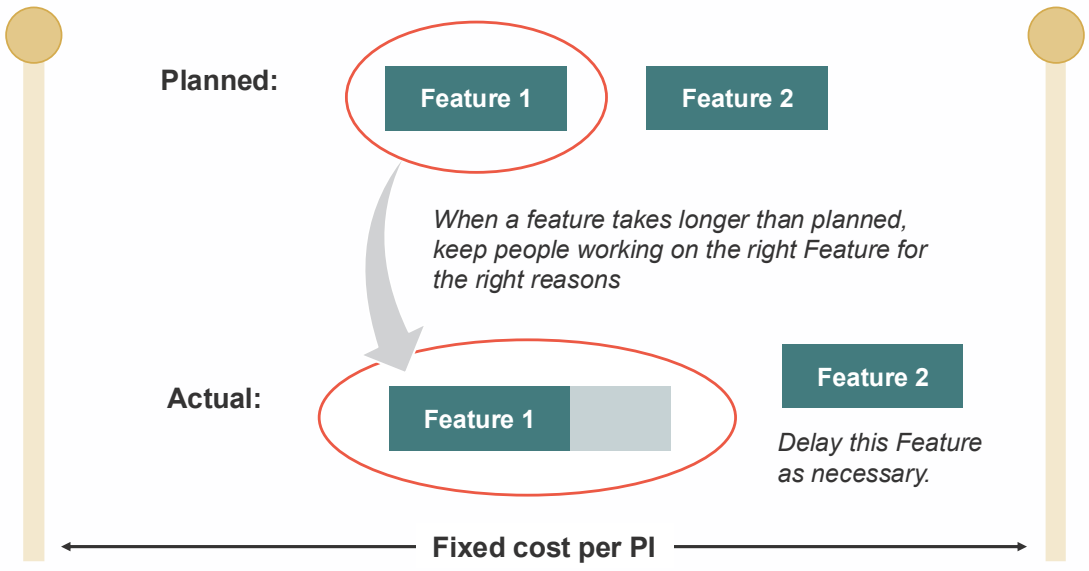
5-29

Budgets are not affected by feature overruns or changing priorities



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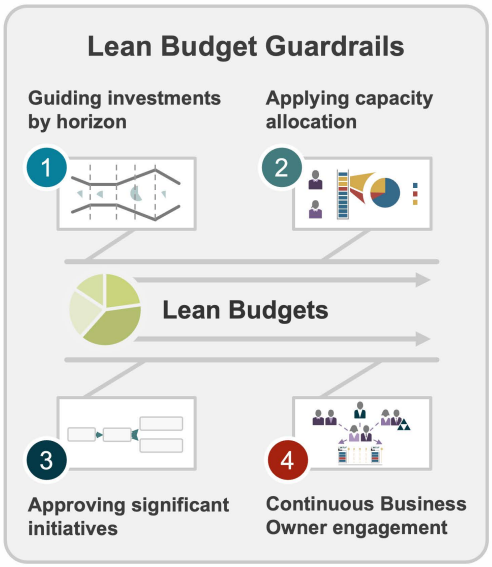
Each portfolio establishes a set of budget guardrails

Guardrails describe portfolio-level budgeting, spending, and governance policies that:

1. Ensure the mix of investments balance near-term opportunities with long-term strategy and growth
2. Balance the backlog of new features with the need to continuously invest in the architectural runway
3. Ensure large, significant investments (epics) are approved appropriately
4. Actively engage Business Owners to ensure the alignment of ART and portfolio priorities



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5.6 Establishing portfolio flow

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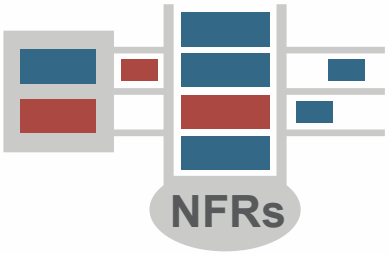
Manage epic flow with the portfolio kanban

- Makes largest business initiatives visible
- Brings structure to analysis and decision-making
- Provides WIP limits to ensure the teams analyze responsibly
- Helps prevent unrealistic expectations
- Helps drive collaboration among the key stakeholders
- Provides a transparent and quantitative basis for economic decision-making



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Epics flow through the portfolio kanban

The portfolio kanban system describes the process states that an epic goes through from the funnel to done.

Funnel	Reviewing	Analyzing	Ready	Implementing		Done
				MVP	Persevere	
All big ideas are captured, such as: • New business opportunities • Cost savings • Marketplace changes • Mergers and acquisitions • Problems with existing Solutions	• Refine understanding of the Epic • Create the Epic hypothesis statement • Preliminary cost estimates and WSJF • WIP limited	• Solution alternatives • Refined cost estimates and WSJF • Define MVP • Create Lean business case • Go/no-go decision • WIP limited	• Epics approved by LPM • Sequenced using WSJF	• Build and evaluate MVP • Pivot or persevere decision made • Pulled by teams	• Affected ARTs or Solution Trains reserve capacity for the Epic • Continue Feature implementation until WSJF determines otherwise	• Done when LPM governance is no longer required
If the idea is below the Epic threshold, move to the appropriate ART Kanban.	Pull when an Epic Owner is available.	Pull when an Epic Owner has capacity.	Pull when approved by LPM.	Pull when train capacity and budget available.	Pull when MVP hypothesis proven true.	Pull when Epic is no longer a portfolio concern.

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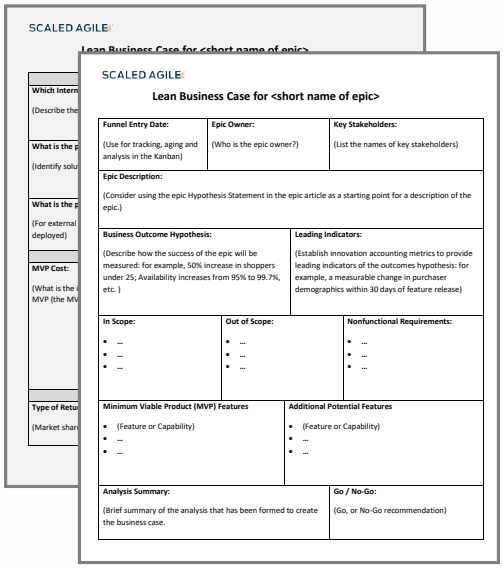
Define the opportunity with a Lean Business Case

- Epics that move into Analyzing need:
  - a Lean business case
  - the definition of a minimum viable product (MVP)
  - an Epic Owner
  - approval by Portfolio Leadership
- The Lean business case is reviewed to make a go/no-go decision for the epic
- Once approved, portfolio epics move to the Ready state of the portfolio kanban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Lean Business Case for <short name of epic>

Funnel Entry Date: (Use for tracking, aging and analysis in the Kanban)	Epic Owner: (Who is the epic owner?)	Key Stakeholders: (List the names of key stakeholders)
Epic Description: (Consider using the epic Hypothesis Statement in the epic article as a starting point for a description of the epic.)		
Business Outcome Hypothesis: (Describe how the success of the epic will be measured: for example, 50% increase in shoppers under 25; Availability increases from 95% to 99.7%, etc.)		Leading Indicators: (Establish innovation accounting metrics to provide leading indicators of the outcomes hypothesis: for example, a measurable change in purchaser demographics within 30 days of feature release)
In Scope: • … • … • …	Out of Scope: • … • … • …	Nonfunctional Requirements: • … • … • …
Minimum Viable Product (MVP) Features • (Feature or Capability) • … • …		Additional Potential Features • (Feature or Capability) • … • …
Analysis Summary: (Brief summary of the analysis that has been formed to create the business case.		Go / No-Go: (Go, or No-Go recommendation)

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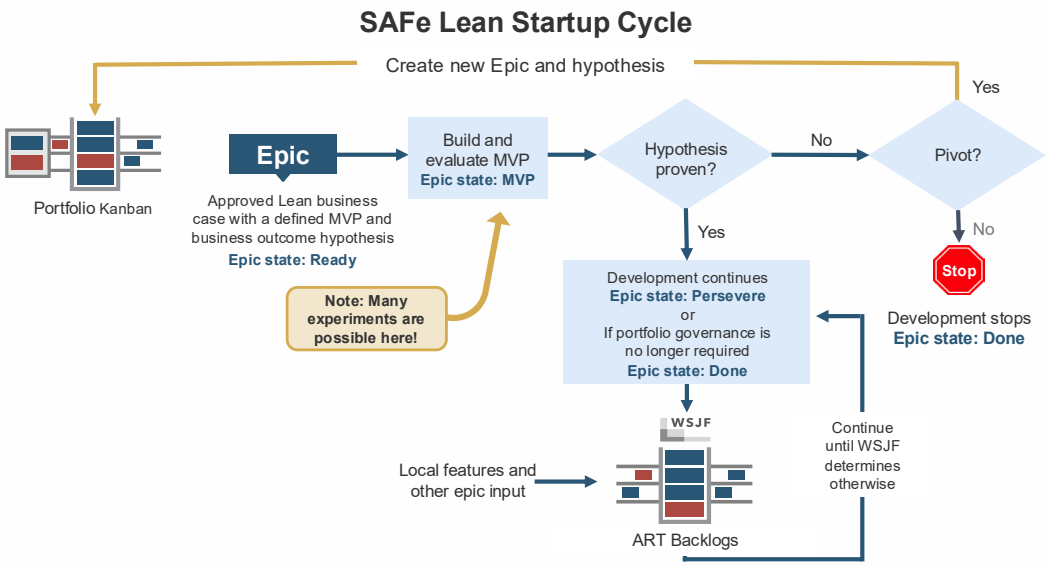
MVPs foster innovation and control scope



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LPM addresses the challenges of a traditional approach

Traditional Approach	Lean-Agile Approach
People organized in functional silos and temporary project teams	People organized in value streams and ARTs
Fund projects and project-cost accounting	Fund value streams
Big up-front, top-down, annual planning and budgeting	Value stream budgets adjusted dynamically
Centralized, unlimited work intake; project overload	Demand managed by portfolio Kanban; decentralized execution by value Streams and ARTs
Overly detailed business cases based on speculative ROI	Lean business cases with MVP
Projects governed by phase gates	Objective measures and milestones based on working solutions

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Lesson review

In this lesson, you:

- Described the purpose and elements of a SAFe portfolio
- Constructed well-written strategic themes
- Employed the portfolio canvas to describe the current and future state
- Created epic hypothesis statements to inform the vision
- Distinguished traditional and Lean budgeting approaches
- Recognized the steps in a Portfolio kanban

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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Lean Portfolio Management," https://framework.scaledagile.com/lean-portfolio-management-discipline/
- "Strategic Themes," https://framework.scaledagile.com/strategic-themes/
- "Portfolio Vision," https://framework.scaledagile.com/portfolio-vision/
- "Lean Budgets," https://framework.scaledagile.com/lean-budgets/
- "Lean Budget Guardrails," https://framework.scaledagile.com/guardrails/
- "Portfolio Backlog," https://framework.scaledagile.com/portfolio-backlog/
- "Epic," https://framework.scaledagile.com/epic/

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Moore, Geoffrey A. *Escape Velocity: Free Your Company's Future from the Pull of the Past.* New York: HarperCollins, 2011. 3.

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Lesson 6

Leading the Change

SAFe® Course - Attending this course gives learners access to the SAFe Agilist exam and related preparation materials.

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Lesson Topics

6.1 Leading by example

6.2 Leading the change

6.3 Empowering leaders with AI

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Learning objectives

At the end of this lesson, you should be able to:

- Explain the skills and behaviors necessary to lead by example
- Discuss techniques for leading successful change
- Recognize the steps in the SAFe Implementation Roadmap
- Identify where AI can most effectively support LPM

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6.1 Leading by example

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Leading by example

- **Insatiable learning:** Depicts how leaders engage in the ongoing, voluntary, and self-motivated pursuit of knowledge and growth, and they encourage and support the same in others
- **Authenticity:** Requires leaders to model desired professional and ethical behaviors
- **Emotional competence:** Describes how leaders identify and manage their emotions and those of others through self-awareness, self-regulation, motivation, empathy, and social skills
- **Courage:** Essential for leaders to guide their organizations through the rapidly changing dynamics of the digital age
- **Growing others:** Encourages leaders to provide the personal, professional, and technical guidance and resources each employee needs to assume increasing levels of responsibility
- **Decentralized decision-making:** Moves the authority for decisions to where the information exists

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Activity: Exploring leadership skills

Prepare: 7 min
Share: 3 min

- **Step 1:** Individually, in your workbook, reflect on the leading by example attributes on the previous slide.
- **Step 2:** Identify one or two of these to focus on in your current context and consider their potential impacts.
- **Step 3:** Be prepared to share with the class.

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Exploring leadership skills

Instructions: Reflect on the leading by example attributes on the previous slide. Identify one to two of these to focus on in your current context and consider their potential impacts.

Notes

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6.2 Leading the change

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Keys to leading successful change

- Dr. John Kotter's 8 accelerators for leading change (contextual and iterative)
- The Big Opportunity is the compelling vision - what results by successfully implementing the change

"The 8-Step Process for Leading Change" by Kotter, Inc.

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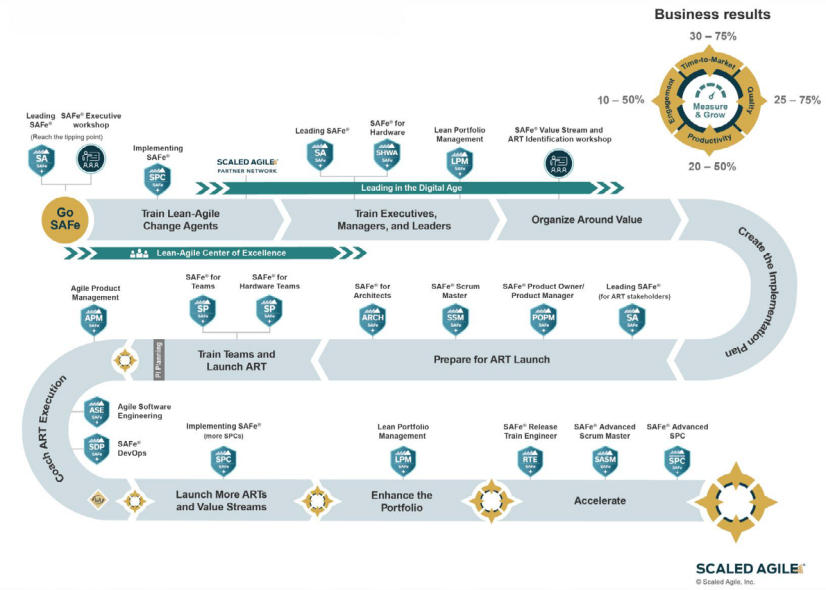
SAFe Implementation Roadmap



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6-9

Train everyone.

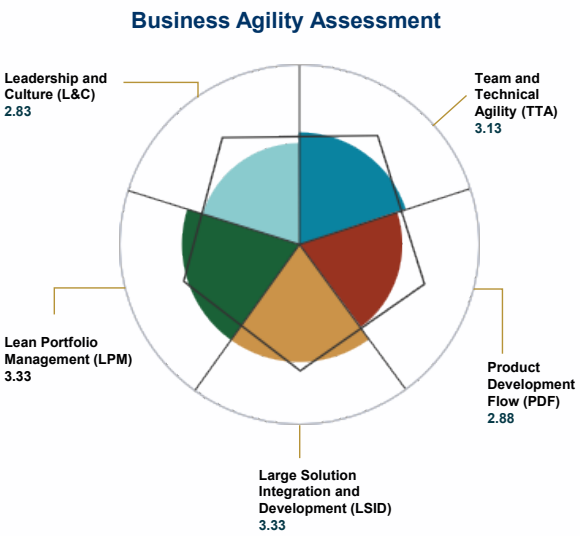
Launch trains.

6-10

Measure your progress toward Business Agility

Measure and Grow is the way each portfolio evaluates its progress toward Business Agility and determines its next improvement steps:

1. Create a high-level summary using the Business Agility assessment
2. Analyze results and identify opportunities to improve



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6-11

6.3 Empowering leaders with AI

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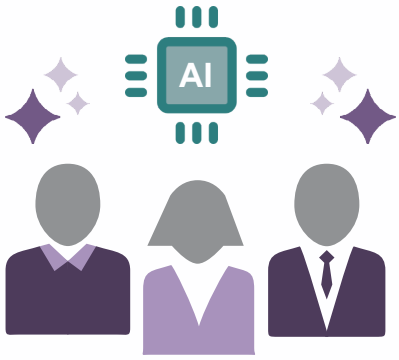
6-12

Why AI matters for portfolio leaders

AI can help leaders see, learn, and respond faster without replacing leadership judgment

- Portfolios operate in environments of high complexity and constant change
- Leaders must synthesize inputs across strategy, investment, flow, and outcomes
- Human judgment remains essential, but signal volume often exceeds human capacity



Portfolio Leadership

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AI as a leadership accelerator

When used responsibly, AI enables portfolio leaders to:

- Learn faster from emerging signals and feedback
- Reduce decision latency while preserving deliberation
- Challenge bias and assumptions more consistently
- Focus leadership attention on vision, tradeoffs, and outcomes

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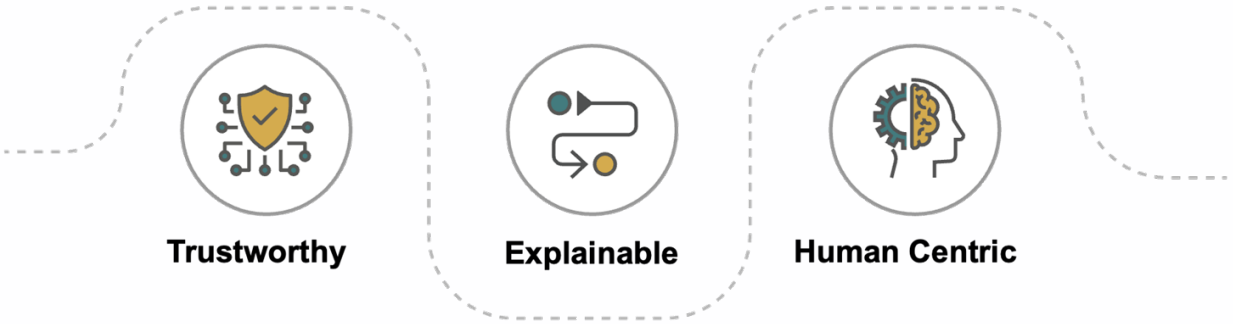
6-15

Aspects of responsible AI



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Trustworthy AI

Trustworthy AI solutions do what they are designed to do reliably, securely, and without exposing confidential data. Attributes include:

- **Privacy:** Ensuring AI systems respect and protect individual and proprietary data
- **Security:** Protecting AI systems from unauthorized access, tampering, or malicious interventions
- **Resilience:** Building AI systems with the ability to withstand adversities, challenges, or attacks
- **Reliability:** Ensuring AI systems consistently perform as expected
- **Accuracy:** Ensuring AI outputs are correct and free from error or hallucinations

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6-17

Explainable AI

Explainable AI solutions provide transparency and documentation that allow humans to understand how their outputs are formulated. Attributes include:

- **Transparency:** Providing a clear, documented understanding of how AI systems operate and make decisions
- **Interpretability:** Making AI decision-making and analysis understandable to humans
- **Accountability:** Holding entities responsible for AI behavior and outcomes

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Human-centric AI

Human-centric AI solutions do no harm to humans, property, or the environment, respecting social norms and values. Attributes include:

- **Safety:** Ensuring AI doesn't pose risks to humans or its environment
- **Fairness:** Ensuring AI systems treat all users and groups equally and impartially
- **Ethics:** Ensuring AI systems uphold society’s moral principles and values
- **Inclusiveness:** Ensuring AI reflects a diverse set of users
- **Sustainability:** Limiting the impact of AI on the environment
- **Compliance:** Ensuring AI adheres to existing laws, regulations, and standards

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Discussion: Responsible AI and Portfolio Leadership

Prepare: 7 min
Share: 5 min

- **Step 1:** Individually, read the following scenario:
  - Portfolio Leadership's AI has analyzed the time-to-market for all value streams. It identifies that the **'Digital Payments'** ART is consistently slowed down by a lack of specialized security engineers.

    To fix this, the AI recommends instantly reassigning four senior security experts from the **Legacy Banking ART** to the **Digital Payments ART** and canceling two upcoming security hires to reduce the labor budget.
- **Step 2:** In your group, evaluate this recommendation using the three aspects of Responsible AI:
  - **Trustworthy:** What data is the AI likely looking at? Does it have the "whole picture"?
  - **Explainable**: What logic or data do you need from the AI to justify this recommendation?
  - **Human-centric:** What happens to the "Legacy" team's morale? What is Portfolio Leadership's role in this shift?
- **Step 3:** Be prepared to share your decision and reasoning with the class.

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6-20

Lesson review

In this lesson, you:

- Explained the skills and behaviors necessary to lead by example
- Discussed techniques for leading successful change
- Recognized the steps in the SAFe Implementation Roadmap
- Identified where AI can most effectively support LPM

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6-21

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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson.

- "Lean-Agile Leadership," https://framework.scaledagile.com/lean-agile-leadership/
- "SAFe Implementation Roadmap," https://framework.scaledagile.com/implementation-roadmap/

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References

Kotter, John. "The 8-Step Process for Leading Change." Kotter, Inc., November 24, 2021. https://www.kotterinc.com/8-step-process-for-leading-change/.

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3. Access your preparation materials and get certified!

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Activity: My Learning Plan

Prepare: 10 min
Share: 5 min

- **Step 1:** Individually, navigate to your learning plan at https://studio.scaledagile.com/
- **Step 2:** In your learning plan, navigate to and complete the course feedback survey
  - Survey results are used to improve courses and help instructors improve. Your feedback will remain anonymous to the instructor.
- **Step 3:** Review the rest of the certification prep materials after completing the survey
- **Step 4:** With your group, make a certification plan to answer the following:
  - How will you fit exam preparation into your schedule?
  - How will you support and hold each other accountable?
  - How will you celebrate your achievement?

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7-3

Video: SAFe Certification Benefits

Duration 3 min

https://share.vidyard.com/watch/ZBApxfn8gYh7W2EpeoEKm4

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7-4

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  Learn proven skills that increase earning potential—SAFe-certified professionals earn more and advance faster.

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  SAFe is the world's leading Agile framework, adopted by Fortune 100 enterprises across industries.

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7-5

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